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College of Information Technology Education

CAPSTONEPROJECTPROPOSAL

***"A Web-Based Integrated School Management System for Tigbauan
Fundamental Learning Center, Inc."***

In partial fulfillment of the requirements in Capstone 1 and Research Subject

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CHAPTER 1

INTRODUCTION

1.1 Background of the Study

Many educational institutions, particularly small private schools located in rural and underserved areas, continue to operate within traditional frameworks where they still heavily rely on outdated, manual processes for managing their most critical daily operations, including the maintenance of comprehensive student records, accurate daily attendance tracking, systematic inventory control, and effective communication between key stakeholders such as administrators, teachers, parents, and learners.

Despite the increasing availability of modern technology and the rapid digitization of various sectors worldwide, these institutions often lag significantly behind in digital adoption, functioning much as they did decades ago while the world around them evolves rapidly toward data-driven efficiency and connectivity.

This delay is rarely due to a lack of desire to improve, innovate, or modernize, but rather stems from practical barriers such as limited and constrained budgets that cannot accommodate expensive commercial software, a scarcity of technical expertise or knowledgeable personnel within the school to manage complex systems, insufficient awareness of available affordable solutions tailored to their specific scale, or deep-seated concerns regarding the complexity of implementing new systems, the difficulty of training staff, and the risk of disrupting established routines.

Consequently, these schools remain dependent on conventional methods—paper

records, handwritten logs, and manual calculations—that result in operational workflows that are not only labor-intensive, slow, and repetitive but also highly susceptible to human error, accidental oversight, and organizational disarray.

The cumulative effect of these inefficiencies creates a heavy administrative burden that diverts valuable time, energy, and intellectual resources away from the core mission of the institution: delivering quality education, facilitating meaningful learning experiences, and supporting the holistic growth and development of every student entrusted to their care.

At Tigbauan Fundamental Learning Center, Inc., the current operational system depends entirely on paper-based files, handwritten logbooks, and basic spreadsheet applications to manage its administrative and academic requirements—a practice that has been in place since its establishment and is deeply embedded in its daily routine, culture, and organizational habits.

While these traditional methods are familiar, straightforward, and require no specialized software or significant hardware investment, they frequently lead to significant operational inefficiencies and administrative burdens that grow heavier, more complex, and more difficult to manage as the school population increases, academic programs expand, and regulatory requirements become more demanding and rigorous.

Every transaction, from the initial enrollment of a learner and the recording of daily attendance to the computation of grades and the eventual release of report cards, requires the physical handling, transfer, and storage of documents, creating a system that is not only slow and cumbersome but also increasingly difficult to manage, scale, and sustain over time.

This heavy reliance on manual documentation creates a fragile operational environment where vital information is scattered across multiple filing cabinets, notebooks, and isolated computer files, making it exceptionally difficult to maintain consistency, ensure data integrity, or guarantee that accurate information is always accessible to those who need it, when they need it.

These manual methods result in extremely slow data retrieval times, where finding a specific student record, past attendance history, or previous year's academic performance can take considerable effort and time, often requiring staff to sort through piles of folders or stacks of logbooks, sometimes spanning several years of operation.

Beyond just being time-consuming and frustrating, there is a heightened and persistent risk of records being misplaced, damaged, or permanently lost due to physical deterioration, filing errors, accidental misplacement, or environmental factors such as humidity, pests, flooding, or fire—risks that directly threaten the institutional memory, historical continuity, and legal compliance of the school.

Furthermore, the preparation of essential documents and reports—such as enrollment summaries, grade sheets, report cards, or attendance reports for submission to government agencies like the Department of Education—is an excessively laborious process.

These tasks require repeated manual computation, transcription, and consolidation of data from different sources, which not only consumes hours of valuable working time that could be better spent on instructional leadership or student support but also frequently leads to calculation errors, inconsistent formatting, missing data fields, and significant delays in decision-making or report submission to regulatory bodies.

When errors inevitably occur, tracing the root cause, verifying the correct information, or correcting the data is a complex, tedious, and frustrating process that further delays operations, disrupts schedules, and affects the reliability and credibility of the school's official records and standing.

Furthermore, communication between teachers and parents remains highly fragmented, uncentralized, and inconsistent, relying largely on ad-hoc methods such as text messages, personal phone calls, informal handwritten notes sent home with students, or occasional group chats on social media platforms.

While these channels work for simple reminders or casual updates, they lack structure, security, privacy, and permanence, often resulting in important messages being overlooked, misunderstood, or forgotten entirely.

This lack of a standardized and structured communication channel often results in delayed notifications regarding important school activities, sudden schedule changes, or emergency announcements that require immediate attention and action.

More critically, it leads to missed or misinterpreted updates on student performance, academic standing, behavioral concerns, or attendance issues—information that is vital for timely intervention, corrective action, and supportive partnership between home and school. There is also no systematic way to document, archive, or retrieve these interactions, meaning important agreements, commitments, or updates can easily be disputed or lost over time.

Consequently, there is an overall poor level of coordination, transparency, and engagement between the school and the home environment—a relationship widely recognized

by educators, researchers, and child development experts as a vital factor in supporting student development, motivation, well-being, and long-term academic success.

Parents often express difficulty in tracking their children's progress in real-time or understanding exactly how their child is performing relative to curriculum standards and learning competencies, as feedback is often limited to quarterly meetings or the distribution of report cards every grading period.

They also face significant challenges in staying updated on school events, requirements, schedules, or financial obligations, leading to confusion, missed deadlines, and occasional non-compliance or misunderstanding regarding school policies and procedures. On the other hand, teachers spend valuable hours outside of class hours following up on inquiries, repeating the same information to different parents individually, or disseminating information through inefficient, one-way channels—time that should instead be dedicated to lesson planning, instructional improvement, professional development, or providing one-on-one academic and emotional support to learners who need it most.

This constant cycle of information chasing and administrative overload disrupts their primary instructional duties and contributes significantly to teacher fatigue, burnout, and job dissatisfaction, which ultimately impacts the quality of teaching and the learning atmosphere within the classroom.

Inventory management and asset monitoring are similarly handled entirely through manual procedures, utilizing paper logs and physical record books to track the movement of classroom supplies, teaching materials, library books, borrowed equipment, furniture, and other school assets.

This approach makes it exceptionally challenging to accurately monitor real-time stock levels, efficiently track item check-outs and returns, identify loss, damage, or wastage immediately, or pinpoint exactly where resources are located within the school premises. Without a centralized system to record and update these transactions, maintaining consistently updated records necessary for accurate budgeting, financial planning, cost analysis, and timely procurement of resources becomes nearly impossible.

Staff members often have to physically inspect storage areas and classrooms just to verify availability or confirm the status of an item—a task that is inefficient, labor-intensive, and highly prone to inaccuracy and oversight. Consequently, the school frequently encounters issues such as unexpected shortages of essential learning materials at critical times like the start of classes, overstocking of non-essential items that tie up limited funds and storage space, and serious difficulties in accounting for assets during internal or external audits conducted by regulatory bodies or funding agencies.

These inventory-related problems strain already limited financial resources, lead to unnecessary waste or repeated purchases of items that are already available but unrecorded, and ultimately disrupt the smooth delivery of educational services and classroom activities, affecting both teachers and students alike.

With the continuous and rapid advancement of information and communication technologies—including cloud computing, web-based applications, mobile computing, and widespread internet connectivity—there is a growing and urgent need for educational institutions, regardless of size or location, to adopt robust digital solutions that can dramatically improve operational efficiency, ensure higher levels of data accuracy and integrity, and provide seamless accessibility to information from any location and at any time.

Modern technology offers the capability to transform administrative work from a heavy burden into a streamlined, automated process, ensuring that data is captured once and reused many times across different functions, significantly reducing redundancy, repetition, and the likelihood of human error.

A well-designed web-based system empowers authorized users—such as school administrators, teaching staff, and registered parents—to access vital data securely and conveniently via internet-connected devices like smartphones, tablets, or laptops, breaking down barriers of distance and time.

At the same time, these systems ensure superior organization through structured databases, standardized workflows, and enhanced security features such as data encryption, automated backups, and role-based access permissions that strictly protect sensitive student and institutional information from unauthorized access, data breaches, or accidental loss, ensuring full compliance with privacy laws and regulations.

The widespread adoption of digital school management systems has already proven remarkably successful in numerous educational institutions across the globe, ranging from large urban public schools in developed nations to small rural academies in emerging economies. In every context, these systems consistently demonstrate substantial improvements in operational performance, transparency, and accountability, creating a more organized and professional environment.

Research and implementation reports indicate that such systems can reduce routine administrative tasks by up to 70%, allowing educators and administrators to redirect their focus, energy, and creativity toward core educational goals such as curriculum development,

enhancing instructional quality, and providing targeted student support services that directly impact learning outcomes.

Furthermore, these systems have been shown to generate markedly higher levels of satisfaction among staff, parents, and students alike, thanks to intuitive user interfaces, transparent information sharing, and reliable system performance that fosters greater trust, confidence, and active participation in the institution's management and community life.

When stakeholders have easy access to accurate information and clear communication channels, the entire school community functions more harmoniously, effectively, and collaboratively.

This study is particularly timely and relevant, as the Philippine government, through the Department of Education (DepEd), continues to strongly promote digital transformation and modernization in school learning platforms and administrative systems via key national initiatives such as the MATATAG Agenda and the Digital Education Act.

These policy frameworks aim to equip all educational institutions—both public and private—with modern tools, resources, and infrastructure necessary for equitable, accessible, inclusive, and high-quality education delivery in the digital age. The national agenda emphasizes that technology integration is no longer merely an option or a luxury for elite institutions, but a necessity for improving governance, ensuring accountability, raising standards, and enhancing the overall quality of basic education across the country.

This mandate specifically includes schools in remote and rural areas where the digital divide remains a significant challenge to be addressed, ensuring that no school, teacher, or learner is left behind in the country's progress toward modernization and global competitiveness.

By developing a customized, web-integrated school management system specifically designed for Tigbauan Fundamental Learning Center, Inc., this research not only aims to contribute meaningfully to the national effort of modernizing basic education but also seeks to directly address the unique operational challenges and constraints faced by small to medium-sized private learning centers in rural communities.

These institutions often face distinct hurdles such as limited financial resources, intermittent or low-speed internet connectivity, lack of dedicated technical support staff, and the critical need for affordable, simple, and user-friendly technology solutions that do not require complex maintenance or specialized training to operate effectively. Unlike generic commercial systems which are often expensive, overly complex, and designed for large-scale institutions, the proposed system is designed with these contextual realities in mind, ensuring that it remains accessible, functional, and sustainable even within limited-resource settings while still delivering the full benefits of professional digital management, security, and scalability.

To comprehensively address these multifaceted challenges and operational gaps identified in the current processes, this study proposes the development and implementation of a fully integrated web-based school management system tailored for Tigbauan Fundamental Learning Center, Inc., which will serve as a unified digital hub for all school operations, bringing together previously disconnected functions into one cohesive platform.

The system is designed to streamline student record management by digitizing personal profiles, academic histories, grade records, and enrollment data, allowing for instant retrieval, secure centralized storage, and easy updating of information while maintaining complete historical records. It automates attendance tracking through a web-based interface that teachers use to mark student presence, absence, or tardiness directly from any connected classroom device, eliminating manual roll-call errors and generating real-time analytical reports that help identify patterns in student attendance and engagement. It also improves inventory monitoring through a centralized digital ledger, automated low-stock alerts, and a full transaction history that simplifies auditing, procurement planning, and resource allocation.

Finally, it enhances communication between teachers and parents via a centralized platform featuring secure messaging, public announcement boards, and academic progress updates, ensuring that everyone stays informed, involved, and connected.

Ultimately, this project aims to create a cohesive digital ecosystem that boosts administrative productivity, reduces operational costs, improves data security and compliance, and fosters stronger collaboration between all stakeholders, leading to better educational outcomes and a more organized, efficient, responsive, and supportive learning environment for every member of the school community.

1.2 Statement of the Problem

This study aims to develop a comprehensive web-based school management system that seamlessly integrates key functionalities—including student records management, realtime attendance monitoring, efficient inventory tracking, grade management, and

streamlined communication tools—into a single, centralized platform to enhance operational efficiency, data accessibility, and collaboration among school administrators, teachers, parents, and students.

At present, many educational institutions continue to rely heavily on manual processes, paper-based documentation, or disjointed software applications to handle their daily administrative and academic operations, creating a fragmented ecosystem that struggles to keep pace with the growing demands of modern education. These conventional methods often result in significant challenges such as unnecessary data duplication, inconsistent record-keeping standards, and extreme difficulty in retrieving or updating information when it is most needed, alongside a consistently increased risk of human error, data loss, or accidental damage to important physical files.

Furthermore, attendance reports are often compiled days or weeks after the fact and shared late or irregularly, making it nearly impossible to promptly identify issues regarding student punctuality or absenteeism, while inventory items such as learning materials, equipment, and supplies are frequently unaccounted for, poorly monitored, or overstocked due to the lack of a systematic tracking mechanism. Additionally, communication between the school and its stakeholders is fragmented across various separate channels—ranging from written notices and text messages to emails and group chats—leading to frequent misunderstandings, missed updates, and a general lack of clarity regarding school activities, policies, or student progress.

Such inefficiencies consume vast amounts of valuable time and financial resources that could otherwise be directed toward improving the quality of education, hinder timely and data-driven decision-making by school leaders, and create substantial barriers to effective

coordination between all parties involved in the learning environment. The absence of a unified system also compromises data security and privacy, as sensitive information is scattered across different locations or formats, and limits accessibility, forcing users to be physically present in specific offices to access necessary records or submit requirements.

Therefore, there is a critical and urgent need for a unified digital solution that consolidates these essential functions into one secure environment, automates routine administrative tasks to reduce workload, ensures high levels of data accuracy and confidentiality, and provides instant access to relevant information regardless of location or time zone. By addressing these existing gaps and limitations in current practices, the proposed system seeks to simplify complex administrative workflows, significantly improve transparency and accountability in school operations, and ultimately foster a more connected, informed, and productive educational community where administrators, teachers, parents, and students can work together more effectively toward academic excellence.

Specifically, this study seeks to answer the following questions:

1. How can student attendance be recorded and monitored more efficiently?
2. How can student records be stored, organized, and retrieved quickly?
3. How can the school effectively manage and monitor inventory and learning materials?
4. How can communication between teachers and parents be improved and centralized?
5. How can reports be generated automatically to reduce manual workload?

1.3 Objectives of the Study

1.3.1 General Objective

To design and develop a comprehensive web-based integrated school management system tailored specifically for Tigbauan Fundamental Learning Center, with the primary objective of significantly enhancing the efficiency, accuracy, and accessibility of all school-related operations and services; this project aims to revolutionize how the institution manages its daily activities and long-term processes, setting a strong, sustainable foundation for digital transformation, operational excellence, and continuous service improvement within the institution.

This proposed system serves as a unified and centralized digital platform that seamlessly consolidates all essential institutional functions—including detailed student records management, real-time attendance monitoring and reporting, systematic inventory tracking of school assets and supplies, grade management, and robust communication tools—effectively replacing the cumbersome manual workflows, voluminous paper-based documentation, and fragmented software solutions currently in use, which often lead to unnecessary delays, data inconsistencies, considerable difficulty in retrieving critical information when it is needed most, and additional workload that drains valuable time and energy from school personnel.

By automating routine administrative tasks such as repetitive data entry, complex report generation, regular record updates, and labor-intensive inventory audits, the system ensures that all information is consistently synchronized across departments, securely stored in a reliable, structured database, and instantly accessible to authorized users with appropriate permissions, thereby drastically reducing human errors, eliminating redundant work processes,

preventing data duplication or permanent loss, and freeing up significant amounts of time and energy for school staff; this allows educators and administrators to focus more fully on core educational goals, student development, instructional quality improvement, and strategic planning rather than being burdened by excessive paperwork, tedious manual verification, and heavy administrative overhead that severely limits productivity and hinders operational growth and innovation.

Furthermore, the platform implements a secure, role-based access control framework, meaning that administrators, teachers, parents, and students each have personalized access to relevant features and data according to their specific needs, responsibilities, and authority levels, ensuring that sensitive information—such as personal details, academic records, and financial data—remains strictly protected against unauthorized access, misuse, or data breaches while simultaneously empowering every stakeholder to retrieve necessary information, submit updates or requests, monitor academic and operational progress, receive urgent or timely announcements, and engage in meaningful, documented collaboration anytime and from anywhere with a stable internet connection, effectively breaking down barriers caused by distance, busy schedules, or limited availability.

Beyond basic operational improvements, the system is designed with scalability and adaptability in mind, capable of supporting the institution's future growth by accommodating increases in enrollment, expanded inventory needs, introduction of new academic programs, and evolving institutional requirements without requiring major overhauls or complete system replacements, while also generating detailed, accurate reports that offer clear, actionable visibility into student performance trends, attendance patterns, inventory status and usage, and overall operational performance.

Ultimately, this integrated system not only streamlines daily operations and strengthens data security and compliance protocols but also provides reliable, real-time insights that support evidence-based, informed decision-making at every level of the institution, foster stronger, more transparent partnerships between the school and families through open engagement and timely, consistent communication, reduce long-term operational costs associated with paper supplies, printing, physical storage, and manual labor, and contribute substantially to raising the overall standard and quality of service delivered by Tigbauan Fundamental Learning Center, thereby positioning the institution to adapt better to modern educational standards, meet and exceed the expectations of its community and regulatory bodies, and serve its learners, staff, and stakeholders with greater excellence, accountability, and reliability for many years to come.

Additionally, the development and implementation of this system are guided by the commitment to create a user-friendly interface designed with intuitive navigation and clear layouts, ensuring that even users with limited technical knowledge or digital literacy can easily navigate, utilize, and maximize the platform's full potential without requiring extensive training or constant technical support. It also aims to establish a cohesive digital ecosystem where every transaction, record modification, and communication is automatically logged and archived, creating an audit trail that enhances transparency, simplifies compliance reporting for regulatory agencies, and preserves the institutional history and data integrity for future reference and evaluation.

Moreover, the system is intended to bridge the gap between home and school by offering parents and guardians a clear window into their children's academic journey, allowing them to view grades, check attendance history, receive notifications regarding school events,

and communicate directly with teachers, thereby encouraging active parental involvement which is widely recognized as a key factor in student success and motivation.

It further seeks to standardize processes across the entire institution, ensuring that policies, procedures, and data formats remain consistent regardless of staff changes or departmental shifts, which in turn creates a stable, predictable, and professional environment that benefits both employees and students alike.

Finally, this initiative represents a vital step toward modernizing the educational framework of Tigbauan Fundamental Learning Center, aligning its operational capabilities with recognized standards of educational management, and ensuring that the institution remains competitive, relevant, and capable of delivering high-quality education in an increasingly digital and interconnected world, while cultivating a culture of innovation, efficiency, and continuous improvement.

1.3.2 Specific Objectives

Specifically, the study aims to:

1. Automate the recording and monitoring of student attendance, enabling real-time data capture and instant updates that transform the way daily student participation is tracked, recorded, and managed within the institution. This feature allows teachers to log attendance through a user-friendly web-based interface accessible from any connected classroom device, significantly reducing the time and effort previously spent on manual checking, copying entries into logbooks, and managing paper-based attendance sheets.

By shifting from traditional roll calls to a streamlined digital process, the system ensures that attendance data is captured accurately at the source and immediately saved to the central

database, eliminating delays in information processing and distribution. It also helps administrators quickly generate comprehensive attendance summaries, statistical breakdowns, and trend analyses whenever required, presenting data in clear formats such as percentages and comparative reports that highlight patterns over days, weeks, months, or entire academic years.

Furthermore, the system allows authorized users—such as class advisers and parents or guardians—to review attendance history directly, ensuring that appropriate and timely intervention measures can be taken when instances of absenteeism, excessive tardiness, or emerging patterns of non-attendance are observed.

By establishing this automated process, the system guarantees accurate, consistent, and reliable record-keeping, reduces errors caused by manual transcription, illegible handwriting, or lost forms, and provides a clear, traceable history of attendance that supports both student welfare monitoring and adherence to institutional compliance standards and government reporting requirements.

2. Centralize and securely store all student records in a comprehensive digital database, systematically organizing critical information that includes personal profiles, enrollment history, class assignments, academic performance records, grade histories, and health information spanning the student's entire period of study from admission to graduation.

This structured repository ensures that every piece of student-related information is logically categorized, properly indexed, easily searchable, and always accessible to those who need it, while being protected through role-based access controls that define exactly what information each user group can view, edit, or manage, effectively preventing unauthorized viewing, accidental modification, or data leakage.

Authorized personnel such as school administrators and class advisers are able to efficiently retrieve, update, or manage records in seconds, whether for routine inquiries, academic evaluations, or official reporting purposes, thereby eliminating the common and costly issues associated with physical file management—such as lost documents, waterdamaged papers, misfiled records, fading ink, or disorganized storage systems that make finding information a tedious and time-consuming task.

Beyond simple storage, this digital archive supports the long-term preservation, integrity, and confidentiality of sensitive student data, while ensuring that information remains intact, retrievable, and usable over many years of operation, creating a reliable institutional memory that supports the school's legacy and accountability.

3. Develop a grade management module that allows subject teachers to record quarterly grades per learner, per subject, per school year, and allows both administrators and parents to view a consolidated, accurate academic performance record. This directly supports the computation of report cards, academic standing evaluations, and the identification of learners who may need additional academic support, replacing the manual tallying of grades on paper class records and report cards.
4. Develop an inventory tracking system capable of managing school resources, covering items including educational equipment, learning materials, textbooks, library resources, office supplies, furniture, and other institutional assets essential for daily operations and effective instruction.

This system actively monitors current stock levels, tracks the movement and distribution of items, and maintains a transaction log of stock changes such as

restocking, adjustments, and disposals, creating a record for every item from the moment it enters the school premises.

Additionally, it features an automated low-stock alert on the inventory dashboard that flags items which have reached a predefined minimum threshold, effectively helping the institution prevent unexpected shortages that could disrupt classes or school activities, reduce unnecessary waste and overstocking of non-essential items, and support better budget allocation and procurement planning.

5. Create a dedicated online communication portal designed for parents, teachers, and administrators, serving as a centralized channel for the exchange of important information, official announcements, and student progress updates throughout the academic year. This feature facilitates transparent, consistent, and timely interaction between all key stakeholders, ensuring that critical information—such as schedule changes, event announcements, or grade updates—reaches the intended audience promptly and reliably, regardless of their location.

The platform supports both direct messaging between parents and teachers regarding specific concerns, and broader announcements visible to all registered users, ensuring that conversations are documented, archived, and retrievable whenever necessary.

6. Generate accurate and automated reports for both academic and administrative use, covering documents such as attendance summaries, grade reports, inventory status reports, and student masterlists.

These reports are produced on request from real-time data stored in the system, formatted consistently to meet institutional requirements. Users have the flexibility to filter reports by

date range, grade level, or subject, and export the underlying data to CSV/Excel format for further analysis, printing, or submission to regulatory bodies. By providing reliable, data-driven insights, these reports serve as a foundation for evidence-based decisionmaking at all levels of management and significantly reduce the manual workload, time consumption, and potential for error traditionally associated with report preparation.

1.4 Scope and Delimitation

The project focuses on the systematic design, full-cycle development, comprehensive testing, and thorough evaluation of a Web-Based Integrated School Management System specifically tailored to address the unique operational requirements, administrative workflows, and academic structures of Tigbauan Fundamental Learning Center, Inc.

The primary objective of this undertaking is to fully digitize the institution's existing manual, paper-based, and spreadsheet-dependent processes—systems that have long been identified as significant sources of operational inefficiency, data redundancy, human error, processing delays, and difficulty in information retrieval. By strategically transitioning these traditional workflows into a secure, dynamic, and fully functional digital environment, the study aims to significantly enhance the accessibility, accuracy, security, and overall integrity of critical institutional data, while drastically improving administrative efficiency, staff productivity, and the quality of service delivery across all departments and functions of the school.

Furthermore, the project seeks to establish a unified, centralized digital platform that seamlessly integrates various essential school functions and serves the distinct yet interconnected needs of all key stakeholders within the school community—including school

administrators, teaching personnel, parents or legal guardians, and duly enrolled students—ensuring that every authorized user has appropriate, role-based access to relevant information and functional tools that align strictly with their specific duties, responsibilities, and level of clearance.

The scope of the study covers the following key features and functionalities, which have been carefully analyzed, selected, designed, and developed to directly resolve the identified operational gaps and challenges currently experienced by the institution:

Student Information Management: This module serves as the core repository and central database foundation of the entire system, encompassing functionalities that cover the complete lifecycle of student data management from admission to graduation or withdrawal. It includes streamlined processes for digital enrollment and admission, allowing for the systematic collection, validation, and secure storage of essential personal details, contact information, and required documentation during the entry phase.

Beyond initial registration, it supports profile updating, enabling authorized administrative staff to modify, correct, and maintain current information whenever necessary to ensure continuous data relevance, accuracy, and completeness. Furthermore, it facilitates the systematic archiving of critical academic records, such as subjects enrolled per term, quarterly and final grades, and attendance history throughout the student's entire stay in the institution.

By structuring, organizing, and consolidating this information into a secure, centralized database, the system ensures that all student records are consistently organized, protected through role-based access controls, and instantly retrievable whenever needed by authorized

personnel only, thereby eliminating the delays, physical storage problems, and risks of loss or damage associated with manual filing systems and the physical handling of paper documents.

Attendance Tracking: This feature is designed to replace traditional logbooks and manual recording methods with a reliable, user-friendly digital logging system that is fully accessible via standard web browsers on desktop computers, tablets, or mobile devices. It allows class advisers and subject teachers to record daily attendance status—clearly indicating whether a learner is present, absent, tardy, or has an approved excused leave—quickly and consistently online.

The functionality goes beyond simple data entry, as the system automatically calculates attendance percentages and generates organized summaries and attendance reports. These outputs are made available to classroom advisors, school administrators, and parents, enabling effective monitoring of student participation, early identification of chronic absenteeism or irregular attendance issues, and data-driven support for student evaluation and intervention programs, while also ensuring that the school maintains precise records necessary for regulatory reporting requirements set by educational agencies and governing bodies.

Grade Management: This feature allows subject teachers to encode quarterly grades for each student, per subject and school year, through a simple web-based grade sheet. Administrators can manage the list of subjects offered per grade level and review all recorded grades across the institution, while parents can view their children's grades and quarterly averages directly through the parent portal.

Inventory Management: Developed to serve as a centralized digital ledger and control mechanism, this module facilitates the efficient tracking and proper accountability of the

school's physical assets, learning resources, and operational supplies. This covers a range of items including library books and reading materials, instructional and teaching aids, classroom furniture, office supplies, and other resources essential for daily academic and administrative operations.

The system streamlines the tracking of the asset lifecycle by featuring a logged history of stock additions, adjustments, and removals, real-time monitoring of current stock levels and quantity on hand, and an automated low-stock indicator on the dashboard. By implementing this digital control mechanism, the system helps prevent loss, misplacement, or unaccounted use of school property, and supports timely procurement planning and resource allocation decisions to prevent shortages or unnecessary overstocking, ultimately contributing to cost reduction and better resource stewardship.

Parent-Teacher Portal: This component provides a dedicated, secure communication platform specifically designed to bridge the existing information gap that often separates the school administration, teaching staff, and the students' homes. The system uses a web-based messaging interface with periodic automatic refresh (AJAX polling) to deliver near real-time updates without requiring specialized server infrastructure, facilitating timely interaction and communication between parents, teachers, and school administrators regardless of distance.

This feature supports private direct messaging regarding individual student concerns or academic performance, as well as the broader dissemination of general announcements.

By housing all communication within this integrated platform, the system ensures that interactions are documented and retrievable for future reference, reducing reliance on informal channels or unregulated group chats.

Report Generation: This reporting module is developed to simplify, accelerate, and standardize the preparation of essential administrative, academic, and operational documents required for internal management, monitoring, and external submission to regulatory bodies. It enables authorized users to generate standardized reports such as Student Masterlists, Attendance Summaries, Grade Reports, and Inventory Status Reports.

Each generated report is formatted to meet institutional requirements and can be viewed directly on-screen for quick review, printed for physical filing and documentation purposes (including saving as PDF through the browser's print function), or exported into CSV/Excel format for further data analysis and computational processing.

This functionality reduces the time and effort spent on manual report preparation and data consolidation, minimizes the risk of computational and transcription errors, and provides reliable, up-to-date documentation that supports precise record-keeping, audit readiness, and facilitates informed, data-driven decision-making by school management and administrative leaders.

In terms of delimitation, the scope of the study is strictly confined to the development, implementation, and evaluation of the modules and features mentioned above, and the system is specifically customized to fit the unique organizational structure, academic programs, curriculum offerings, and operational policies of Tigbauan Fundamental Learning Center, Inc. It does not cover the development of a full-scale Learning Management System (LMS) intended for conducting virtual classes, hosting online lectures, delivering course content, or facilitating digital assignment submissions and online examinations.

Furthermore, the system does not integrate online payment gateways, electronic banking facilities, or process any form of financial transactions; any fee-related information or financial data displayed within the system, if any, is included solely for monitoring, record-keeping, and reference purposes only, and does not serve as a replacement for dedicated accounting or financial management software.

The system also does not implement biometric or QR-code-based attendance authentication; attendance is recorded manually by the teacher through the web interface, consistent with the institution's current infrastructure and budget constraints. Additionally, the operation of the system is inherently dependent on the availability of stable and consistent internet connectivity and compatibility with modern web browsers; technical support or troubleshooting regarding hardware malfunctions, power interruptions, server downtime, or issues related to the internet service provider (ISP) and network infrastructure are explicitly outside the scope and coverage of this research and development project.

1.5 Significance of the Study

The development and implementation of this Integrated School Management System hold significant value and provide substantial, far-reaching benefits to the following academic stakeholders, serving as a vital instrument in the modernization of school operations while fostering a more connected, efficient, transparent, and productive educational environment that fully aligns with current standards of quality education, administrative excellence, and institutional governance:

School Administrators: This system serves as a powerful, comprehensive

management tool that offers school administrators a centralized, intuitive dashboard designed to oversee, monitor, and control all institutional operations, resources, and data flows from a single, accessible interface accessible anywhere and anytime.

It ensures a high level of data integrity, consistency, and security by effectively eliminating data redundancy, conflicting records, and human errors that are commonly prevalent in manual, paper-based, or fragmented systems, thereby providing a solid, reliable foundation for informed, evidence-based, and strategic decision-making.

By having instant, real-time access to critical data regarding student enrollment records, academic standing and performance metrics, daily attendance trends and statistics, and inventory status, administrators can efficiently monitor the school's overall performance, enforce policies and regulations consistently and effectively across all departments, and plan for strategic improvements, infrastructure development, or resource allocation with much greater accuracy, confidence, and foresight.

Furthermore, the availability of automated, standardized reports significantly simplifies compliance with government regulatory requirements, reduces the heavy workload associated with auditing, accreditation, and documentation, and empowers management to lead the institution with greater transparency, full accountability, and sustained operational excellence.

Teaching Faculty: For teachers and academic personnel, this technological development significantly reduces the heavy administrative burden and clerical workload traditionally associated with their professional roles by automating repetitive, timeconsuming, and routine tasks such as daily attendance recording, computation and tabulation of grades, and the preparation of various academic and administrative reports.

By minimizing the time and effort spent on extensive paperwork, manual calculations, and physical record-keeping, the system allows educators to redirect their valuable focus, energy, and attention toward their primary and most important responsibilities: enhancing instructional quality, refining and improving teaching strategies, and supporting the holistic development of every learner. With simplified, streamlined processes and the ability for instant retrieval of complete student information—including academic history, attendance patterns, and relevant personal details—teachers can better understand the specific needs, strengths, and challenges of their learners, dedicate more quality time to lesson planning, curriculum development, and interactive classroom instruction, and provide more timely, meaningful feedback and targeted intervention, ultimately elevating the overall quality of education delivered to the learners and fostering a more supportive learning environment.

Non-Teaching Staff: This system benefits non-teaching personnel and support staff by streamlining resource management and logistical operations through digitized inventory tracking and asset monitoring features. It reduces the occurrence of manual errors, data discrepancies, and the volume of physical paperwork, ledgers, and filing that typically slows down daily operations and complicates record retrieval.

The platform facilitates systematic, accurate monitoring of all school assets, consumable supplies, equipment, and materials, simplifies the recording of stock additions and removals, and automates alerts regarding low stock levels.

Consequently, this helps staff maintain optimal and adequate inventory levels to prevent disruptive shortages or unnecessary overstocking, prevent loss or misuse of school resources, and speed up documentation, auditing, and reporting workflows required by the

administration and external agencies. This leads to a more organized supply chain, optimized utilization of budget and funds, and a smoother, more efficient operational flow that effectively supports the entire institution's functions and goals.

Parents and Guardians: The study strengthens and enhances the vital school-home partnership by providing parents and guardians with direct, secure access to essential information regarding their children's education, including daily attendance records, academic progress and grading details, and schedules of upcoming school activities. Through a dedicated, user-friendly, and secure parent portal, the system facilitates secure, direct communication channels with teachers and school administrators, effectively removing barriers, delays, and uncertainties often found in traditional notification methods such as newsletters or occasional meetings.

This constant and transparent access keeps parents consistently well-informed, actively engaged, and deeply involved in their child's educational journey, strengthens the level of transparency, confidence, and trust between the school and the families it serves, and allows for timely collaboration and constructive dialogue in addressing any concerns, challenges, or specific needs regarding the student's welfare, conduct, and academic performance, thereby creating a unified support system for the learner.

Students: While students are not direct operators or managers of all administrative modules within the system, they stand to gain significant indirect yet highly impactful benefits from the improved efficiency, accuracy, transparency, and organization brought about by the implementation of this technology.

It ensures that their personal profiles, enrollment details, and academic records are accurately maintained, securely stored, properly monitored, and consistently updated throughout their entire stay in the institution, significantly reducing the risk of errors, omissions, or data loss that could potentially affect their academic standing, certification, or transfer credentials.

Additionally, because the system supports better inventory management, timely procurement, and strategic resource allocation, students are assured of a learning environment supported by well-managed facilities, readily available learning materials, and organized resources necessary for effective instruction.

Furthermore, the enhanced communication loop established between school and home ensures that their academic growth, attendance regularity, and behavioral development are closely tracked, guided, and supported by both educators and parents, creating a more organized, responsive, and supportive educational setting that prioritizes their academic success, safety, well-being, and overall development.

Future Researchers: This research and its resulting system serve as valuable reference material, comprehensive documentation, and a solid foundational basis for further studies, academic inquiries, innovations, and technological developments related to educational technology, school management systems, and web-based applications within the broader field of information technology and educational management.

The detailed design principles, system architecture, specific features, development methodologies, testing results, and evaluation findings derived from the development and assessment of this project can effectively guide future researchers, software developers, and

students in understanding the unique operational requirements of small to medium-sized educational institutions, overcoming common challenges related to limited-resource settings, and developing more advanced, scalable, and specialized systems tailored to specific educational contexts.

It provides a tested and validated framework that can be expanded, modified, enhanced, or integrated with emerging technologies—such as artificial intelligence, data analytics, or mobile computing—to explore new functionalities, improve service delivery models, and further enhance the efficiency, reach, and quality of educational administration and governance in similar contexts.

1.6 Definition of Terms

Integrated School Management System – refers to a comprehensive software suite and digital infrastructure specifically designed to digitize, unify, and streamline the wide range of academic, administrative, and support processes required to operate an educational institution effectively.

It consolidates diverse functions—including the management of student records, attendance monitoring, grade management, logistical inventory control, and stakeholder communication—into a single, cohesive, and efficient platform that serves as the central nervous system of the school's daily operations.

This system functions by connecting different departments, user roles, and stakeholders—such as administrators, teachers, students, and parents—within a shared digital environment, ensuring seamless data flow, consistency in information, and coordinated

workflows that effectively replace traditional, fragmented methods involving separate databases, manual files, and disconnected tools.

By bringing all essential functionalities together in one accessible system, it significantly simplifies complex management tasks, improves the speed and quality of service delivery, enhances transparency and accountability, and provides the necessary foundation to support the institution's strategic transition toward a fully digital, paperless, and highly organized educational environment where information is accurate, secure, and readily available to those who need it.

Real-time Messaging – is defined as a communication feature embedded within the system that enables the timely delivery and receipt of text-based information, notifications, and updates between authorized users. In this system, this is implemented through AJAX polling, where the browser periodically checks the server for new messages every few seconds, allowing conversations to update without requiring the page to be manually reloaded.

This capability ensures that school announcements, updates regarding schedule changes, and important reminders regarding events and deadlines are transmitted and received promptly, allowing for timely responses and action from recipients.

Unlike traditional delayed messaging systems, physical notice boards, or third-party applications, this feature facilitates two-way interaction and a dedicated channel for exchanging important inquiries, clarifications, and general information, all secured within the institutional system and without dependence on external services or additional communication costs.

Attendance Tracking – represents the systematic digital process of accurately recording, continuously monitoring, and formally reporting the presence, absence, punctuality, and participation status of students within the school premises or during scheduled learning sessions.

Built to replace outdated manual logbooks, paper registers, or roll-call methods, this functionality utilizes a secure, web-based interface accessible by teachers to capture attendance data and update central records, while automatically storing historical data for long-term reference and analysis.

Beyond simple logging, this feature allows administrators and authorized personnel to generate reports, identify recurring patterns of absenteeism or tardiness, and improve institutional accountability and compliance. It serves as a vital tool for early intervention, simplifies the creation of official attendance summaries required for regulatory compliance, and provides a clear, verifiable record that helps evaluate student engagement and instructional regularity.

Grade Management – refers to the process of digitally recording, storing, and reporting a learner's quarterly and final academic grades per subject and school year. It replaces the manual class record book and hand-computed report card, allowing subject teachers to encode grades directly into the system and allowing administrators and parents to view a consolidated, accurate, and consistently formatted academic record for every enrolled learner.

Inventory Management – constitutes a systematic, data-driven approach to overseeing, controlling, and optimizing the school's ecosystem of physical assets, consumable supplies, educational materials, and furniture.

It encompasses the lifecycle of managing resources—from initial recording and storage, to issuance, adjustment, and eventual disposal or replacement—ensuring that every item belonging to the institution is properly accounted for, tracked, and available exactly when and where it is needed. Through organized recording, monitoring features, and automated low-stock alerts, this system is designed to prevent disruptive shortages that hinder instruction, reduce waste or loss due to misplacement, and optimize budgeting by providing clear visibility into stock levels.

By maintaining a centralized, real-time record of all school property, this process helps minimize shrinkage, ensures the efficient and equitable utilization of limited resources, and generates historical data that aids school leaders in planning future purchases and sustaining the quality of facilities, tools, and learning materials essential for the educational process.

Student Records Management – refers to the comprehensive and systematic collection, secure storage, regular updating, and efficient retrieval of all relevant personal, academic, and administrative information regarding learners throughout their entire journey within the institution.

This includes detailed personal identification data, enrollment history, class assignments, academic performance metrics, and attendance logs. The primary objective of this management process is to ensure that highly sensitive data is securely stored, logically

organized, and accessible only to authorized personnel according to strict privacy protocols, thereby protecting student confidentiality.

At the same time, it guarantees that accurate, complete, and up-to-date student profiles are instantly available for administrative processing, academic evaluation, curriculum planning, or official reporting purposes. This system effectively eliminates common issues related to misplaced, damaged, or disorganized physical files, reduces redundancy and errors, and allows educators and administrators to track student progress holistically, ensuring that every learner's educational history is preserved and accessible for decision-making and support services.

Report Generation – describes the process of compiling, aggregating, and transforming raw data stored within the system into structured, readable, and professionally formatted documents such as summaries, detailed lists, and statistical breakdowns.

This functionality allows authorized users to produce a range of essential administrative and academic outputs—including attendance summaries, grade reports, inventory status reports, and student master lists—formatted in usable file types such as CSV/Excel, or printed/saved as PDF through the browser.

By automating what was previously a labor-intensive and error-prone manual task, this feature saves time and effort, reduces the risk of calculation mistakes or data inconsistencies, and provides reliable, verifiable documentation that supports regulatory compliance requirements, internal and external evaluation processes, and evidence-based decision-making. It serves as the primary mechanism for translating raw data into actionable insights, enabling school leaders to assess performance, identify trends, and allocate resources wisely.

Role-Based Access Control (RBAC) – is a security framework used throughout the system that restricts access to specific pages, features, and data strictly according to the authenticated user's assigned role (Administrator, Teacher, or Parent). Every request to a protected page is checked against the user's role on the server before any data is shown, ensuring that, for example, a parent account cannot view or modify teacher-only functions such as grade encoding, and a teacher account cannot access administrator-only functions such as user management.

Audit Trail – refers to the chronological, timestamped log the system keeps of significant actions performed by users, such as logging in, adding or editing a student record, adjusting inventory stock, or creating a user account. Each entry records who performed the action, what was done, and when, supporting accountability, troubleshooting, and compliance reporting.

1.7 Theoretical Framework

This study is anchored on three complementary theories that collectively justify both the decision to computerize the operations of Tigbauan Fundamental Learning Center, Inc. and the specific design choices made in the proposed system.

The Technology Acceptance Model (TAM), originally advanced by Davis (1989), posits that an individual's willingness to use a new information system is primarily determined by two beliefs: perceived usefulness, or the degree to which a person believes that using the system will enhance their job performance, and perceived ease of use, or the degree to which a person believes that using the system will be free of effort. TAM is directly relevant to this study because the intended end users — the school administrator, teachers, and parents — are, for the most part, not information technology specialists. If the proposed Web-Based Integrated

School Management System is to replace long-standing manual routines, it must be demonstrably useful in reducing clerical workload and demonstrably easy to learn without extensive technical training. Every interface decision described in Chapter 3, from the simplified dashboard layouts to the use of familiar terminology drawn from the institution's existing forms, is guided by this theoretical concern for perceived usefulness and ease of use.

Second, the study draws on Rogers' (2003) Diffusion of Innovation Theory, which explains how, why, and at what rate new ideas and technologies spread through a social system. Rogers identifies five attributes that influence the adoption of an innovation: relative advantage, compatibility, complexity, trialability, and observability. Applied to this study, the proposed system is designed to have a clear relative advantage over paper-based recordkeeping (faster retrieval, reduced duplication of effort), to be compatible with the school's existing academic calendar, grading periods, and reporting requirements, to minimize complexity through role-based dashboards, and to allow trialability by permitting a phased rollout — beginning with student records and attendance before extending to grades, inventory, and messaging — so that administrators and teachers can observe tangible benefits before the system is used institution-wide.

Third, the study is informed by General Systems Theory, particularly as applied to information systems through the classic Input-Process-Output (IPO) model. This theory conceives of an organization as a set of interrelated components working together to transform inputs into meaningful outputs, and it underlies the systems analysis and design approach adopted in Chapter 3. Viewing the school as a system whose components — enrollment, attendance, grading, inventory, and communication — are interdependent rather than isolated activities allows the researchers to design a single, integrated platform rather than a set of

disconnected tools, ensuring that data entered once (for example, a student's section assignment) is available wherever it is needed (attendance sheets, grade sheets, and reports) without re-encoding.

Taken together, these three theories provide the conceptual justification for (a) why an integrated digital system is an appropriate response to the problems described in Section 1.2, (b) how the system should be introduced to maximize the likelihood of sustained adoption, and (c) how its internal components should be structured to function as a coherent whole rather than a collection of separate modules.

1.8 Conceptual Framework

The conceptual framework of the study is expressed through the Input-ProcessOutput (IPO) model, which illustrates the logical flow of the research and development effort from the identification of institutional needs to the delivery of a working, evaluated system.

On the input side, the study relies on four categories of information: the existing manual records and forms currently used by the institution (enrollment forms, attendance logbooks, grade sheets, and inventory ledgers); the requirements and pain points gathered directly from the administrator, teachers, and parents through interviews and surveys; the relevant related literature and studies reviewed in Chapter 2, which establish good practice in the design of school information systems, attendance modules, inventory modules, and communication portals; and the technical constraints of the institution, including its available hardware, internet connectivity, and budget for hosting.

The process component covers the actual development effort, structured according to

the Agile phases described in Section 3.2 — requirements analysis, system design, iterative development, testing, and deployment — together with the data-gathering and evaluation procedures described in Sections 3.5 through 3.7. It is during this stage that the raw inputs are transformed, through systems analysis and software engineering practice, into database schemas, user interfaces, and application logic.

The output of the study is the Web-Based Integrated School Management System itself, together with the accompanying documentation — the data dictionary, user manuals, and test results — and the intended institutional benefits: faster and more accurate recordkeeping, reliable attendance and grade computation, better-controlled inventory, and a functioning communication channel between the school and parents. A feedback loop connects the output back to the process, since post-deployment observations and user feedback (see Section 3.11) are expected to inform subsequent refinements of the system beyond the scope of Capstone 1.

Component	Description
Input	Manual records and forms; stakeholder requirements gathered via interview and survey; reviewed literature and studies; institutional technical constraints (hardware, connectivity, budget).
Process	Requirements analysis, system design, iterative Agile development, functional and usability testing, deployment and evaluation.
Output	Web-based integrated school management system; data dictionary and user documentation; measurable improvements in record accuracy, retrieval speed, and communication.

Table 1.1 — Input-Process-Output Framework of the Study

1.9 Assumptions of the Study

The study proceeds on the basis of the following assumptions, which the researchers consider reasonable given the context of Tigbauan Fundamental Learning Center, Inc.:

- **Availability of infrastructure.** The institution has, or can obtain, sufficient internet connectivity and at least one computer or mobile device per key user role (administrator and teachers) to operate the proposed system.
- **Willingness to transition.** The administration, teaching staff, and parents are generally willing to shift from manual, paper-based processes to a digital system, provided that the system is demonstrably easier to use and clearly beneficial.

- **Accuracy of existing records.** The manual records currently maintained by the institution (enrollment lists, attendance logs, grade sheets) are substantially accurate and can serve as a reliable basis for initial data migration into the new system.
- **Stability of institutional processes.** The core academic processes of the school — the grading periods, subject offerings, and attendance policy — will remain relatively stable during the development period, so that the system requirements defined in Chapter 3 remain valid through implementation.
- **Scope limited to Capstone 1 and 2.** Full-scale, multi-year deployment, integration with government reporting systems such as the Learner Information System (LIS), and mobile application development are recognized as valuable extensions but are treated as recommendations for future work rather than commitments of the present study.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

2.1 School Management & Student Information Systems

These systems function as comprehensive centralized repositories that consolidate all student-related data, effectively streamlining a wide spectrum of administrative and academic operations including enrollment procedures, grade computation and maintenance, report generation, curriculum mapping, and compliance documentation, thereby enabling educational institutions to operate with significantly higher levels of efficiency, drastically reducing dependence on voluminous paper records, and ensuring that accurate, up-to-date information is readily accessible to authorized users such as school administrators, teachers, parents, and the students themselves at any given time.

By bringing together data that was previously scattered across different departments, filing systems, or separate software tools, these integrated platforms eliminate the confusion,

delays, and inconsistencies that often arise from fragmented information management, creating instead a single, reliable source of truth that aligns all institutional activities and ensures everyone works from the same accurate dataset.

They automate repetitive and time-consuming tasks such as data entry, form processing, record updates, and the preparation of regulatory reports, which not only minimizes the risk of human error and data duplication but also frees up valuable time for school personnel to focus on core educational responsibilities, student support, and program improvement.

Furthermore, with built-in security measures and role-based access controls, the systems guarantee that sensitive information—including personal details, academic history, and health records—remains protected against unauthorized access, misuse, or loss, while simultaneously allowing authorized stakeholders to retrieve relevant details, track academic progress, review attendance records, or communicate updates securely from any location with internet connectivity, breaking down geographical barriers and enabling remote engagement.

This seamless availability of reliable data fosters greater transparency and accountability within the institution, strengthens collaboration between the school and families by keeping parents actively informed and involved, and provides school leaders with the accurate, real-time insights necessary for informed decision-making, strategic planning, resource allocation, and continuous enhancement of the overall quality of education and services provided.

Additionally, these systems serve as a dynamic operational backbone that interconnects aspects of school life, from admissions and registration to graduation, creating a continuous digital thread that follows each learner throughout their entire educational journey,

ensuring that no critical detail is lost or overlooked during transitions between grade levels or academic years.

By standardizing data formats, coding structures, and reporting frameworks, these platforms also ensure that all institutional information adheres to established educational policies and industry best practices, thereby simplifying external audits, accreditation evaluations, and submissions to educational governing bodies, while reducing the administrative burden associated with meeting strict compliance deadlines and

documentation standards.

Ultimately, School Management and Student Information Systems act as more than just data storage or administrative tools; they are enablers of educational excellence that transform how schools operate, collaborate, and serve their communities, driving digital transformation, fostering a culture of continuous improvement, and ensuring that educational institutions remain agile, competitive, and equipped to deliver high-quality learning experiences.

iSercan: A Web-Based Student Information System (iSercan, 2019)

This research details the systematic development, comprehensive design, and successful implementation of a robust web-based portal engineered specifically to provide enrolled students and their parents or guardians with secure, round-the-clock access to essential academic details and educational records.

The platform delivers key functionalities including grade updates and progress tracking, class schedules, and transparent fee payment monitoring, all while placing a strong emphasis on data security, reliable system performance, and cross-platform compatibility across different

devices. These technical features and functional capabilities are strategically aimed at minimizing administrative bottlenecks and processing delays, significantly reducing institutional reliance on traditional paper documentation and manual filing systems, and fostering continuous, smooth, and transparent information sharing between the school administration, faculty members, and the home environment.

The study thoroughly documents the entire software development lifecycle—from initial requirements gathering, stakeholder interviews, and system analysis, through architectural design, coding, and testing phases, to final deployment and user training—ensuring that every feature and module is built to align precisely with actual institutional needs and user expectations derived from direct observation and empirical feedback.

It highlights how the platform effectively organizes, consolidates, and centralizes critical data that was previously fragmented, dispersed across various offices, or stored in separate physical files and databases, making it significantly easier for school administrators and staff to manage vast volumes of student records, generate accurate and standardized reports required for compliance or planning, and update information instantly across the system without the risk of duplication, inconsistency, or human error.

The research underscores how the system's structured role-based access controls—restricting data visibility according to user type—and data encryption protocols work together to protect sensitive personal, academic, and financial information from unauthorized access, external data breaches, or accidental loss, thereby addressing common concerns regarding privacy, confidentiality, and data integrity that often accompany the transition to digital school management systems.

By streamlining communication channels, standardizing information presentation, and making vital details readily available in a user-friendly format, the platform successfully bridges the traditional information gap between the school and families, encourages active and consistent parental involvement, and helps build a more collaborative, informed, and supportive educational community (iSercan, 2019).

The findings presented demonstrate that this web-based solution not only improves operational efficiency and reduces the repetitive workload of administrative and teaching staff by automating routine tasks and record-keeping processes but also enhances the overall quality of service delivery, institutional accountability, and operational transparency within the educational setting, serving as a reliable model and valuable reference for modernizing school information management practices and digital transformation initiatives in similar academic environments (iSercan, 2019).

Systems Analysis of the Philippine Basic Education Ecosystem (Cuyegkeng, 2026)

This insightful working paper, developed in partnership with the Second Congressional Commission on Education (EDCOM II), undertakes a thorough examination of the broader digital ecosystem within the Philippine Department of Education, utilizing a systems thinking approach to holistically map out and analyze the complex interrelationships, functional dependencies, and operational workflows among various stakeholders, institutional processes, and existing technological infrastructures currently in place across the sector.

Through this rigorous assessment, the research clearly identifies the urgent and critical necessity for the adoption and implementation of fully integrated data management frameworks and interoperable systems capable of effectively breaking down persistent information silos,

standardizing diverse data formats, and ensuring a seamless, secure, and consistent flow of information across different levels, regions, and administrative units of the entire education sector.

Such integrated technological systems are envisioned to provide substantial, evidence-based support to school leadership and management teams at all levels, significantly enhancing both the quality and speed of strategic decision-making processes by delivering accurate, real-time insights and comprehensive data analytics. These capabilities enable administrators to monitor institutional progress and performance effectively, allocate limited resources with maximum efficiency and transparency, and align local institutional goals, practices, and curricular offerings with national education standards, government policies, and long-term development agendas such as the MATATAG framework and the broader digital transformation roadmap.

Furthermore, the study emphasizes that these modern technological solutions play a pivotal role in significantly strengthening communication channels and collaboration networks, successfully bridging existing gaps and fostering meaningful, continuous engagement across educators, administrators, learners, families, and community partners to create a more connected, inclusive, and supportive educational environment.

It further highlights that improved, timely, and equitable access to accurate, reliable, and actionable educational data serves as a key intervention point and a foundational strategy to drive better learning outcomes for all students, strengthen governance and accountability mechanisms within schools, and ensure more effective, responsive, and uniform implementation of education policies and programs nationwide.

Ultimately, the paper asserts that establishing a cohesive digital infrastructure is indispensable for addressing systemic challenges, reducing administrative burdens, and advancing the overall modernization and transformation of the Philippine basic education system to meet global standards and future demands (Cuyegkeng, 2026).

The research further delves into the historical and structural roots of fragmentation within the current system, explaining how decades of decentralized operations, independent reporting mechanisms, and the adoption of disparate, non-compatible software tools by different divisions and schools have created significant bottlenecks and inconsistencies in data interpretation that hinder effective governance and policy execution across the archipelago. It outlines the essential characteristics that integrated systems must possess to succeed within the Philippine education landscape, including high adaptability to varying levels of digital readiness across regions, robust offline capabilities for areas with limited connectivity, strict adherence to national data privacy laws and security protocols, and usercentric designs that accommodate the diverse technical skills of personnel working in both urban centers and remote rural communities.

The study envisions a future where every school in the Philippines, regardless of location or size, is connected to a secure, reliable, and integrated digital ecosystem that supports every aspect of its operations, concluding that investing in this cohesive digital infrastructure is a strategic imperative that will serve as the backbone of educational reform (Cuyegkeng, 2026).

Making Education Manageable: School Management Systems and Data-Driven Classrooms (Taylor & Francis, 2025)

This critical analysis investigates the discursive construction and evolving functional role of School Management Information Systems (MIS), closely examining how these

technological tools are reshaping the landscape of modern education by prioritizing data tracking capabilities, performance benchmarking, and analytics measures across all levels of institutional operation.

It explores how such systems influence and redefine the definition of educational quality, accountability standards, and success metrics, demonstrating how data-driven insights are increasingly embedded into every aspect of school life—from curriculum planning and student assessment to attendance monitoring, teacher evaluation, and resource distribution.

The study further delves into the impacts these technologies have on daily classroom dynamics and instructional strategies, illustrating how educators utilize real-time information to tailor teaching interventions, monitor individual and group progress, and address specific learning needs, enabling a more personalized and responsive approach to education.

Simultaneously, it highlights how these systems transform the priorities, operational workflows, and resource allocation strategies of school administration, shifting them from reactive, manual-based practices to more strategic, evidence-based decision-making processes.

Beyond immediate practical utility, the analysis engages with the broader implications and ongoing academic debates surrounding digital integration in educational settings, raising important discussions about the necessary balance between leveraging data to improve outcomes and preserving pedagogical autonomy and the human-centered values that lie at the heart of teaching and learning.

By evaluating both the transformative benefits—such as enhanced transparency, improved efficiency, and targeted support—and the potential challenges associated with a heavily data-centric approach, this work provides a balanced perspective on how these integrated systems are shaping the future of teaching, learning, and educational governance (Taylor & Francis, 2025).

The analysis also critically examines the complexities inherent in this digital evolution, raising thoughtful concerns regarding the risk of over-reliance on quantitative metrics, the possibility of reducing complex educational experiences to mere numbers, and the challenges of ensuring data privacy, security, and ethical use in an environment where information is constantly collected and analyzed. It underscores the importance of ensuring that technology serves as an enabler and support system rather than a rigid control mechanism, concluding that the true value and success of School Management Information Systems depend entirely on how they are designed, implemented, and governed (Taylor & Francis, 2025).

A Web-Based Expert System for Student Advising (Philippine E-Journals, n.d.)

This study, originating from research conducted at the Technological Institute of the Philippines (TIP), concentrates on the systematic design, development, and functional testing of an automated web-based expert system created specifically to handle the detail-oriented, labor-intensive nature of the student academic advising process.

It highlights how such a technological innovation can significantly reduce the processing duration of critical tasks—such as curriculum evaluation and prerequisite validation—from a matter of days or weeks under manual operations to near-instantaneous results within the digital environment, effectively eliminating unnecessary delays and

mitigating the risks of oversight, inconsistency, and human error frequently associated with traditional, person-dependent advising methods.

By formally encoding institutional policies and curriculum requirements directly into its decision-making framework, the system is capable of delivering personalized, accurate, consistent, and unbiased academic guidance and course recommendations efficiently and reliably, regardless of the volume of students being served.

Furthermore, the research demonstrates that this digital solution reduces the administrative burden traditionally placed on academic advisors, allowing them to redirect their focus toward meaningful mentorship and instructional improvement rather than routine data verification. It also ensures that students receive clear, accessible, and timely information regarding their academic standing, empowering them to make well-informed decisions about their education.

Ultimately, the study establishes that implementing such an expert system not only streamlines institutional workflows and improves operational efficiency but also enhances the overall quality and reliability of student support services, promotes fairness and objectivity in academic processes, and contributes to a more organized, standardized, and responsive educational environment (Philippine E-Journals, n.d.).

The study also emphasizes the value of standardization across the advising process, noting that in traditional settings, guidance quality can vary significantly depending on an individual advisor's familiarity with complex rules or interpretation of requirements; by contrast, this digital solution applies rules uniformly and objectively to every student, ensuring

equitable treatment and guaranteeing that every learner receives consistent, accurate information (Philippine E-Journals, n.d.).

Cloud-Based Student Information System for Educational Institutions (Al-Mutairi et al., 2023)

Al-Mutairi and colleagues (2023) conducted an in-depth exploration regarding the application and strategic integration of cloud computing technology in the design, development, and deployment of modern student information systems, placing a strong emphasis on core advantages such as scalability, cost-efficiency, and remote accessibility for all authorized users regardless of their location or device.

Their study demonstrated that cloud-based solutions allow educational institutions of varying sizes—from small rural learning centers to large universities—to implement secure and fully functional data management frameworks without the heavy burden of significant capital investment in physical servers, complex hardware setups, or the need for extensive on-site technical maintenance and specialized IT personnel.

By leveraging a centralized, internet-based infrastructure, the system facilitates realtime data synchronization across different departments or campuses, ensuring that critical information such as student records, grade submissions, and attendance logs remains consistent and up-to-date across the organization, effectively eliminating data discrepancies and redundant entries that often plague fragmented, on-premise systems.

It also enables secure, role-based access to relevant information and functionalities from any location and on any internet-enabled device—a feature that proved highly beneficial during periods of remote or hybrid learning, where continuous connectivity and uninterrupted access

to academic resources and administrative tools were critical to sustaining educational services without disruption.

Additionally, the research highlights how this technological approach strengthens data protection and institutional resilience through automated backups and encryption protocols managed by professional service providers, which safeguards sensitive information against loss, damage, or unauthorized access more reliably than conventional manual or local server systems.

Ultimately, the authors conclude that adopting cloud-based student information systems streamlines administrative workflows, reduces long-term operational expenses and maintenance costs, and enhances the overall reliability, availability, and quality of educational services delivered to stakeholders, effectively making modern, efficient, and secure data management accessible, affordable, and sustainable for educational institutions regardless of their scale or available resources (Al-Mutairi et al., 2023).

The study further underscores the importance of data integrity and continuity, noting that because information is stored in geographically distributed, redundant data centers managed by professional providers, an institution is fully protected against catastrophic events such as hardware failure or local system crashes, a level of protection that is difficult to achieve with traditional in-house setups without substantial additional investment (AlMutairi et al., 2023).

Data Integration and Interoperability in School Management Systems (Chen & Huang, 2024)

Chen and Huang (2024) conducted an in-depth investigation into both the technical complexities and organizational challenges associated with merging disparate, standalone, and often isolated school management systems commonly adopted by educational institutions.

Their research highlights that the lack of proper interoperability, standardization, and compatibility between different software solutions or internal functional modules frequently leads to the formation of restrictive data silos—isolated repositories where critical information remains confined within specific departments or administrative functions and cannot be easily accessed, shared, or utilized across the broader institution.

This fragmentation of information flow, they observed, results in significant operational inefficiencies, including the necessity for repetitive and redundant data entry tasks, inconsistent record-keeping practices, and substantial difficulties in consolidating information to generate comprehensive reports that accurately reflect the true overall state of school operations.

In response to these issues, their research proposed a standardized framework for seamless data exchange and system integration, specifically designed to establish common protocols and consistent communication standards that enable different functional modules—such as student enrollment, academic grading, attendance monitoring, and inventory management—to interact and function together as a cohesive, unified ecosystem rather than separate disjointed tools.

By ensuring that data flows smoothly and consistently across all areas of the organization, this integration framework eliminates unnecessary duplication of effort, enhances

overall data accuracy and reliability, and provides school administrators with a complete, holistic view of institutional performance, resource utilization, and educational outcomes.

Ultimately, the study demonstrates that achieving high levels of integration and interoperability within school management systems streamlines daily administrative workflows and improves the quality and speed of service delivery to all stakeholders, while empowering educational leaders with the accurate, real-time information needed for effective data analysis and strategic planning (Chen & Huang, 2024).

The authors also discuss the transformative impact of such integration on institutional analytics, noting that when data from academic performance, attendance patterns, and resource usage is combined into a unified database, it unlocks analytical capabilities that were previously unattainable, allowing leaders to identify correlations—such as the relationship between resource allocation and student achievement—that provide deep insights for improvement and strategic investment (Chen & Huang, 2024).

2.2 Attendance Monitoring Systems

Automated attendance solutions leverage technological tools such as web-based interfaces, mobile applications, and, in more advanced implementations, QR code scanning or biometric verification, to improve the precision and accuracy of daily attendance recordkeeping, reduce the administrative workload and labor costs associated with manual roll calls and repetitive data entry, and minimize the occurrence of common human errors and recording discrepancies.

These systems are engineered to capture attendance details promptly, removing much of the subjectivity and oversight found in traditional methods, while simultaneously generating

structured, reliable datasets that reflect student engagement patterns, participation trends, and behavioral habits over time.

By replacing traditional paper-based logbooks and sign-in sheets with digital tools, educational institutions ensure that attendance data is captured at the point of entry and recorded within the system's database, reducing the data discrepancies, missed entries, or processing delays that typically arise from manual compilation and filing.

These systems automatically store, organize, and archive attendance logs within a centralized database, allowing for integration with other school management modules such as student information records and reporting tools, ensuring that attendance data directly contributes to a more comprehensive view of each learner's academic journey and overall standing in the institution.

Beyond record-keeping and compliance, the accumulated historical data serves as a valuable analytical resource, helping educators and institutional leaders identify recurring trends, such as chronic absenteeism or correlations between attendance and academic performance, which can inform the design of targeted support programs and strategic decision-making aimed at improving overall attendance rates and boosting learning outcomes.

The ability to generate customizable reports—ranging from daily summaries to detailed monthly analytics—allows school administrators to monitor compliance with national education policies and quickly identify classes or groups where participation rates are dropping, enabling proactive measures before minor issues escalate into serious concerns.

Moreover, by automating the process of computing attendance percentages and compiling data for report cards, these systems reduce the likelihood of computational errors

and save time that would otherwise be spent manually tallying records, allowing staff to redirect their time and energy toward educational support and student welfare initiatives.

Optimized QR Code-Based Authentication System for Attendance Management (Abirami & Rohann, 2026)

Published in 2026, this detailed technical paper puts forward an optimized and secure QR code system enhanced with encryption techniques and dynamically generated unique session tokens, designed explicitly to counteract common fraudulent attendance practices such as "buddy punching," unauthorized access, or the duplication and misuse of attendance credentials by third parties.

Recognizing the significant security limitations inherent in standard static QR codes—which remain unchanged for long periods and can therefore be easily captured or shared among students to mark attendance without being physically present—the proposed framework introduces a mechanism where every code generated is uniquely time-sensitive and cryptographically protected against tampering, rendering invalid any attempts to reuse or forge them for illegitimate purposes.

Developed with the goal of enhancing the reliability and security of attendance records particularly within large-scale academic institutions where monitoring large student populations poses significant logistical challenges, the system enables instantaneous verification of user identity and automated logging of attendance data directly into a centralized, secure database, reducing processing delays and transcription errors so often found in traditional paper-based or semi-digital attendance recording methods.

The study clearly demonstrates how this optimized technological solution strengthens institutional accountability, streamlines daily administrative workflows, and provides accurate,

trustworthy data that supports the enforcement of institutional policies regarding attendance and academic compliance (Abirami & Rohann, 2026). While this level of QRbased authentication is beyond the current scope of the proposed system for Tigbauan Fundamental Learning Center, it establishes a useful direction for future enhancement should the institution later expand toward automated attendance verification.

Smart School Attendance: Virtual Solution for Secondary Level Schools (Chávez-Saldaña & Guevara-Ramos, 2026)

This comprehensive study introduces a QR-code-based attendance platform designed to facilitate efficient student registration, daily attendance tracking, report generation, and active parental monitoring through automated notifications and secure access to attendance history. By bridging the traditional information gap between school administration and the home environment, the system ensures that parents are continuously informed about their child's presence, absence, or tardiness.

Significantly, the research specifically addresses the challenges of technology implementation within educational environments characterized by low or unstable internet connectivity, limited digital infrastructure, or lack of technical support—conditions prevalent in many developing regions and rural areas—acknowledging that standard fully clouddependent systems are often impractical in such settings.

To overcome these barriers, the authors developed a hybrid data architecture that supports offline data capture and local storage, allowing teachers to record attendance seamlessly even without an active internet connection, with an automated synchronization mechanism that updates the central database once connectivity is restored.

This context-aware design demonstrates that it is possible to implement modern, transparent attendance systems that improve administrative efficiency and strengthen collaboration among stakeholders even in challenging operational contexts (Chávez-Saldaña & Guevara-Ramos, 2026). This offline-first design consideration is a relevant reference point for Tigbauan Fundamental Learning Center given its rural setting, even though the current proposed system assumes stable connectivity is available during school hours, consistent with the scope defined in Chapter 1.

A Students Attendance System Using QR Code (Masalha & Hirzallah, 2014)

This foundational research explores the practical application and educational benefits of assigning unique QR codes to every enrolled student, which are scanned by instructors at the start of every learning session to automate the attendance logging process and replace conventional manual procedures such as calling names or marking checklists.

The authors noted that this scanning-based method significantly reduces the disruption to class time typically associated with traditional roll calls, allowing educators to dedicate more attention to instruction. The study highlights that this technological approach allows for the immediate generation, secure digital storage, and easy retrieval of accurate attendance records, eliminating common issues inherent to manual systems such as illegible handwriting or lost documents.

Ultimately, the research establishes that QR code-based attendance systems serve as a reliable, cost-effective, and user-friendly solution that enhances overall administrative efficiency and supports better monitoring and accountability regarding student participation (Masalha & Hirzallah, 2014).

Biometric-Based Attendance System: Enhancing Security and Accuracy (Singh & Singh, 2022)

Singh and Singh (2022) conducted a comparative analysis of technologies used for attendance monitoring, focusing on fingerprint and facial recognition biometric systems within educational settings. Their findings indicated that while the initial setup and hardware costs of biometric solutions are comparatively higher than simpler alternatives such as QR codes or password-based logins, these systems offer superior levels of authentication security and consistent long-term reliability.

A key advantage is that biometric verification relies on unique physiological traits that cannot be transferred or duplicated, which eliminates the risk of credential sharing or proxy attendance ("buddy punching"). The authors also noted that such systems can be configured with automated alerts regarding patterns of absenteeism, serving as an earlywarning tool for educators to implement targeted intervention strategies.

Ultimately, the research concludes that biometric-based attendance systems deliver long-term value by guaranteeing data integrity and strengthening institutional accountability, though the higher upfront investment makes them more suited to larger institutions with adequate budgets than to small rural learning centers such as Tigbauan Fundamental Learning Center (Singh & Singh, 2022).

Mobile-Based Attendance Monitoring with Geolocation Features (Garcia & Santos, 2023)

Set within the operational and geographical context of the Philippines, Garcia and Santos (2023) designed and tested a mobile attendance system that combines QR code scanning with geolocation verification to ensure that attendance is only logged when students or teaching

personnel are physically present within predefined school premises, preventing off-site or fraudulent attendance marking.

Conducted within public high schools, their study revealed that the implementation of this digital solution reduced administrative workload and time spent on attendance-related paperwork by up to 60%, and significantly improved the speed, accuracy, and compliance of reporting required by Schools Division Offices.

The platform also functions as a communication hub, sending instant notifications to parents regarding their child's arrival, departure, or unexpected absence, strengthening the connection and trust between educational institutions and families (Garcia & Santos, 2023).

2.3 School Inventory & Asset Management

Effective inventory and asset management systems serve as essential operational components for educational institutions, as they ensure that critical school supplies, instructional materials, technological equipment, library resources, furniture, and other tangible assets are continuously monitored, accurately tracked, and fully accounted for throughout their entire lifecycle—from initial recording and storage, through issuance and utilization, to eventual disposal or replacement.

By maintaining a centralized, real-time record of assets and stocks across different departments, classrooms, and storage facilities, these systems reduce the risks of loss, damage, misplacement, or unauthorized use, while eliminating the data discrepancies and operational inefficiencies commonly associated with manual stock-taking and paper-based record keeping.

Furthermore, they facilitate optimized procurement strategies and smarter budget utilization by generating data-driven insights regarding consumption patterns, current stock

levels, and reorder requirements, enabling administrators to purchase only what is necessary, at the right time and quantity, thereby avoiding both costly overstocking and disruptive shortages.

Ultimately, the implementation of robust inventory and asset management practices contributes directly to greater financial accountability, stricter compliance with auditing standards, improved quality of service delivery, and the establishment of a more organized, secure, and resource-efficient learning environment that sustains the school's educational mission.

Web-Based Inventory Management System for Educational Training (Syamsi et al., 2026)

This paper discusses the design and integration of predictive mathematical models—specifically the Autoregressive Integrated Moving Average (ARIMA) model for demand forecasting, and the Economic Order Quantity (EOQ) model for determining optimal stock levels and reorder points—into a functional, web-based inventory management system.

Custom-designed for practical application within educational training centers and technical-vocational institutions, where specialized tools and consumable training materials constitute a significant portion of operational expenses, the system is intelligently built to maintain ideal stock levels that align with actual usage rates and curriculum requirements.

By leveraging these analytical frameworks, the platform assists administrators in minimizing unnecessary holding costs, reducing excess stock accumulation and material waste, and preventing critical material shortages that could interrupt instructional activities. This dual-purpose solution enhances institutional efficiency and financial accountability while enriching the overall learning experience by bridging classroom instruction with modern industry-

standard practices (Syamsi et al., 2026). While the proposed system for Tigbauan Fundamental Learning Center does not implement ARIMA/EOQ forecasting given the institution's smaller scale, this study informed the decision to include a simple low-stock threshold alert as an appropriately-scoped equivalent.

Inventory Management Practices and Perceived Financial Performance (Baliuag University, n.d.)

This assessment provides a detailed overview of fundamental inventory management principles and industry best practices as applied within a range of Philippine-based enterprises, government agencies, and non-profit organizations, highlighting the critical role of quantitative techniques such as demand forecasting and scientific reorder point determination in maintaining a balance between minimizing waste and maximizing efficient resource utilization.

Significantly, the study draws clear parallels between commercial operations and educational administration, establishing that these structured principles are directly transferable and equally essential to institutional asset management contexts in schools, colleges, and learning centers. By adopting these scientific approaches, educational institutions can effectively streamline procurement processes and prevent both supply shortages and unnecessary financial burden caused by overstocking.

Ultimately, the study asserts that applying these management standards ensures that every financial investment and physical asset is properly accounted for and strategically utilized to directly support educational programs and student support services, thereby elevating the overall quality of institutional governance (Baliuag University, n.d.).

Design-Based Research to Develop an Inventory Management Tool (Brajets, 2025)

This study employs the Design-Based Research (DBR) framework to guide the iterative development, testing, and validation of a specialized inventory management tool tailored to address the operational contexts of educational environments. By following this research-oriented approach, the development process ensures that every feature is grounded in real-world user needs and institutional goals.

The resulting system is capable of tracking the movement, location, and condition of diverse assets throughout their lifecycle, while generating data-driven reports that support strategic planning, budgeting, and procurement activities. The research demonstrates how implementing such a structured tool leads to improved decision-making and higher operational efficiency, particularly within the common budget constraints characteristic of smaller organizational settings such as individual schools or rural learning centers, concluding that even modest, well-designed inventory systems can deliver substantial institutional value (Brajets, 2025).

Asset Tracking and Management in Educational Institutions Using RFID Technology (Khan et al., 2023)

Khan and associates (2023) conducted an in-depth exploration into the application of Radio Frequency Identification (RFID) technology for tracking and managing fixed assets—including computers, laboratory apparatus, furniture, and library collections—within university settings where monitoring high volumes of valuable items poses significant logistical challenges.

Their research demonstrated that RFID-enabled systems revolutionize traditional inventory procedures by allowing for faster, non-intrusive stock-taking, where hundreds of

individually tagged items can be scanned simultaneously, drastically reducing the time and labor costs previously spent on manual counting and verification.

The study also highlighted how the system facilitates better maintenance scheduling by automatically logging usage data and repair records, ultimately leading to a reduction in accidental asset loss or misplacement while extending the functional lifespan of institutional infrastructure. The authors concluded that automated asset tracking significantly strengthens institutional accountability and simplifies compliance with government auditing standards (Khan et al., 2023). RFID hardware investment of this scale is beyond the resource constraints of Tigbauan Fundamental Learning Center as identified in Chapter 1, reinforcing the study's decision to instead implement a lower-cost, transaction-log-based digital inventory ledger.

Optimizing School Supply Chains through Digital Inventory Systems (Dela Cruz & Reyes, 2024)

Focusing specifically on the operational realities faced by public schools throughout the Philippines, Dela Cruz and Reyes (2024) conducted a comprehensive analysis to determine how the systematic implementation of digital inventory systems can enhance the management, distribution, and utilization of essential learning materials and educational resources.

Their findings revealed that continued reliance on manual inventory methods often leads to processing delays, data inaccuracies, and severe stock imbalances where some schools face acute shortages while others hold excess, unused supplies representing wasted investment. By digitizing inventory management procedures, schools were able to reduce lead times for stock replenishment and enable more precise, real-time monitoring of stock levels and consumption rates.

The authors concluded that transitioning to integrated digital inventory solutions creates a more responsive, transparent, and efficient educational supply chain, ensuring that limited public funds are utilized wisely and that every learner and teacher has timely, reliable access to necessary resources (Dela Cruz & Reyes, 2024).

2.4 Parent-Teacher Portals & Communication

Digital communication tools and dedicated online portals are developed and implemented with the primary aim of effectively bridging the traditional information gap between the home and school environment by enabling continuous, secure two-way exchange of information and meaningful engagement among parents, guardians, teachers, and school administrators.

These integrated platforms serve as centralized hubs where parents and guardians can conveniently access essential details regarding their child's educational journey, such as academic progress reports and attendance records, while also providing direct, structured channels to communicate inquiries, raise concerns, or schedule consultations with teachers and school officials.

For educational institutions and management teams, these technologies offer significant strategic benefits including improved transparency in operations, enhanced trust between families and school staff, and greater levels of active parental involvement—which has been widely documented and linked in educational research to better student performance, higher attendance rates, and improved behavioral outcomes.

However, while the advantages are substantial, their widespread adoption also introduces new considerations and potential pressures for educators and school personnel. Key

challenges frequently identified include the additional burden of increased administrative workload required to maintain digital records, the growing societal expectation of constant availability outside standard working hours, and the difficulty of managing information overload caused by frequent notifications.

Furthermore, there are critical considerations regarding equitable access and inclusivity, as disparities in digital literacy levels, device ownership, or reliability of internet connectivity among families may inadvertently create new divides or exclude certain groups from full participation. As such, successful and sustainable implementation requires careful strategic planning, the establishment of clear guidelines regarding communication boundaries and response times, and a balanced approach that maximizes the engagement benefits for students and families while protecting the professional well-being of educators.

***Does Parent-Teacher Communication Truly Boost Students' School Performance?
(MPRA, 2025)***

This critical study offers a balanced, evidence-based evaluation of the actual impact and practical effectiveness of digital engagement platforms and parent-teacher portals within educational settings, demonstrating through empirical data that while these systems often contribute positively to an increased perception of improved student performance and engagement through features such as status updates and grade notifications, there is a significant downside and potential risk when usage becomes excessive, unregulated, or poorly structured.

The research highlights that without clearly defined boundaries and agreed-upon response frameworks, the constant flow of inquiries and messages from stakeholders can result in observable "negative spillover effects" that ultimately undermine the very educational

quality and outcomes these tools were designed to support. These adverse outcomes include heightened levels of teacher burnout, professional fatigue, and a reduction in the available time and cognitive focus necessary for essential core tasks such as lesson planning and individualized student support.

Ultimately, the study concludes that when digital communication channels become overwhelming or are left unmanaged, they inadvertently detract significantly from a teacher's available time and attention during actual classroom instruction, potentially negating the intended benefits of enhanced connectivity (MPRA, 2025). This finding directly informed the decision to limit the proposed system's messaging feature to a straightforward, asynchronous message board rather than a constant-notification chat platform, reducing the risk of the negative spillover effects this study describes.

The Influence of Teachers' Outreach to Parents on Students' Academic Success (SAJHE, 2025)

This comprehensive research investigates the efficacy of various digital platforms as mediums for the structured dissemination of student progress reports and behavioral feedback, as well as strategic tools designed for the cultivation of deeper parental engagement in the learning process.

Through extensive analysis of educational performance data, the study confirms that consistent, active, and informative communication between teachers and parents strongly correlates with improved student outcomes across multiple indicators, including higher academic achievement, better attendance records, and increased learning motivation.

However, the research also offers a balanced perspective by highlighting significant limitations and contextual challenges inherent in modern outreach strategies, noting that

persistent socio-economic barriers, differing levels of digital literacy, and unequal access to reliable internet connectivity or suitable digital devices can still hinder the realization of truly inclusive and equitable communication for all families.

Consequently, the authors emphasize the necessity of adopting flexible hybrid approaches that seamlessly combine modern digital outreach with traditional, accessible methods such as printed notices or face-to-face meetings, ensuring that every parent or guardian, regardless of their economic background or technical capability, remains fully informed and capable of participating in their child's education (SAJHE, 2025).

Digital Parent-Teacher Communication: Benefits and Challenges in K-12 Education (Lee & Ward, 2023)

Lee and Ward (2023) conducted a comprehensive review of existing literature and case studies regarding the implementation and impact of digital communication channels between schools, teachers, and families within the K-12 educational context.

Their extensive synthesis of data indicates that when these platforms are utilized effectively and with clear purpose, they foster a true partnership approach to education where parents and guardians are consistently better informed about classroom activities and individual student needs, thereby equipping them with the necessary knowledge to actively support learning objectives at home.

However, the authors also identified significant challenges associated with unregulated or poorly managed digital communication, including the emergence of an "always-on" culture that blurs professional boundaries and leads to educator fatigue, the inherent risk of misinterpreting tone in text-based messages, and the persistent issue of the digital divide.

Based on these findings, the study suggests that for successful and sustainable implementation, educational institutions must establish clear communication guidelines, define appropriate response times, and provide necessary training for both staff and parents to ensure these tools serve to enhance rather than complicate the educational process (Lee & Ward, 2023).

Parental Engagement and Academic Achievement: The Role of Technology (Al-Mashrafi, 2024)

Al-Mashrafi (2024) conducted a thorough investigation into the complex relationship between technology-mediated parental engagement and student academic performance, systematically analyzing performance data and interaction patterns across diverse educational settings to understand how digital tools influence the quality of home-school collaboration.

The findings clearly demonstrated that engagement facilitated through structured digital platforms is positively associated with higher academic grades, improved attendance records, and better overall student behavior; however, the research also revealed a crucial distinction: the quality, relevance, and constructiveness of the communication exchanged matter far more than the sheer frequency or volume of messages sent.

The study established that simply providing regular updates or generic notifications does not automatically translate to better educational results; rather, the information shared must be meaningful, specific, and explicitly focused on guiding parents regarding concrete ways they can support learning at home. Consequently, the study emphasizes that digital platforms should be designed to facilitate genuine two-way dialogue rather than functioning merely as passive one-way channels for broadcasting general information (Al-Mashrafi, 2024).

2.5 Legal and Regulatory Basis

Beyond the technical and pedagogical literature reviewed in the preceding sections, the design of the proposed system is also anchored on relevant Philippine laws and administrative issuances governing the handling of student and institutional data.

Republic Act No. 10173, or the Data Privacy Act of 2012, requires that any entity that collects and processes personal information — which, in the case of a school, includes the names, birthdates, addresses, and academic records of minors — implement reasonable and appropriate organizational, physical, and technical security measures to protect that information against accidental or unlawful destruction, alteration, and disclosure. This law directly informs the access-control, authentication, and audit-logging features described in Sections 3.3 and 3.4, since the proposed system will store personal data belonging to minors and their families and must therefore restrict access strictly according to user role.

Republic Act No. 8792, the Electronic Commerce Act of 2000, establishes the legal recognition of electronic documents, electronic signatures, and electronic transactions in the Philippines, giving digitally generated records — such as system-generated report cards and attendance summaries — the same legal weight as their paper counterparts, provided that reliable methods are used to maintain their integrity. This provides the legal basis for the institution to eventually rely on system-generated reports as official records rather than treating them merely as a convenience layer over paper files.

Finally, relevant Department of Education issuances on school-based management and learner information systems — including the general policy framework for the K to 12 Basic Education Program — establish the recordkeeping obligations that private learning centers such as Tigbauan Fundamental Learning Center, Inc. must satisfy, particularly with respect to

the retention of learner permanent records, submission of enrollment and completion data, and the maintenance of accurate class and attendance registers. The proposed system's reporting module (Section 3.8) is designed with these obligations in mind, so that the exports it produces can support, rather than complicate, the institution's compliance with such requirements.

2.6 Synthesis of Related Literature and Studies

The literature and studies reviewed in Sections 2.1 through 2.4 converge on several consistent findings that directly shape the design of the proposed system.

First, across school management, attendance, inventory, and communication systems alike, the reviewed works consistently report that digitization reduces the time required to retrieve and process records compared to manual methods, and that this efficiency gain is one of the primary motivations cited by institutions that adopt such systems (Al-Mutairi et al., 2023; Chen & Huang, 2024). This finding directly supports the core premise of the present study, articulated in Section 1.1, that Tigbauan Fundamental Learning Center, Inc. stands to gain significant operational efficiency from a similar transition.

Second, several of the reviewed studies on attendance monitoring highlight a persistent trade-off between the sophistication of the identification mechanism used (biometric, QR-code based, or RFID-based) and its practicality in resource-constrained settings (Masalha & Hirzallah, 2014; Singh & Singh, 2022; Khan et al., 2023). Given that Tigbauan Fundamental Learning Center, Inc. is a small private institution operating on a limited budget, the present study deliberately favors a simpler, teacher-encoded attendance mechanism over biometric or RFID alternatives, prioritizing low hardware cost and ease of maintenance over the marginal gains in automation that such technologies would provide.

Third, the literature on parent-teacher communication portals consistently finds that two-way, system-mediated communication channels improve parental engagement and, in turn, correlate with improved student outcomes, but only when the channel is simple enough for parents with limited digital literacy to use consistently (Lee & Ward, 2023; Al-Mashrafi, 2024). This finding directly informed the decision, described in Section 3.8, to design the parent-facing dashboard around a small number of clearly labeled views (attendance, grades, messages) rather than a feature-dense interface.

Fourth, the reviewed inventory management literature demonstrates that even simple, rule-based digital inventory tracking — without the more advanced forecasting techniques such as ARIMA or Economic Order Quantity modeling used in larger institutions (Syamsi et al., 2026) — already yields substantial improvements over manual ledger systems in terms of stock visibility and reduction of loss (Brajets, 2025; Dela Cruz & Reyes, 2024). Because the scale of Tigbauan Fundamental Learning Center, Inc.’s inventory needs does not currently justify the complexity of demand-forecasting algorithms, the present study adopts a rulebased reorder-level approach, leaving predictive inventory analytics as a possible future enhancement.

Finally, while the reviewed works collectively address attendance, grading, inventory, or communication in isolation, few of the local and foreign studies examined describe a single system that integrates all four functions within one authentication and data model, particularly in the context of a small, resource-constrained Philippine private school. This is the specific gap that the present study seeks to address: rather than proposing another singlepurpose tool, the researchers propose an integrated platform in which a student’s identity, section assignment, attendance, grades, and household communication are all governed by one consistent data

model, reducing the duplication of encoding effort and the risk of inconsistency that arises when separate systems are used side by side.

CHAPTER 3

METHODOLOGY

3.1 Research Design

The study employed a developmental research design, a methodological approach specifically focused on the systematic design, rigorous development, and comprehensive evaluation of the proposed technological application.

This framework is highly suitable and appropriate for this research as it provides a structured, step-by-step guide for creating an effective, functional solution while simultaneously ensuring that the final product meets essential standards of functionality, usability, technical reliability, and relevance to the specific needs and contexts of the target users.

The research followed a well-defined framework that encompasses three primary, interconnected phases: the planning and design phase, where extensive requirements gathering, detailed system analysis, and conceptualization of features and architecture were conducted based on identified institutional needs, user expectations, and existing operational challenges; the development phase, involving the actual coding, database creation, interface design, and integration of various functional components to construct a working prototype of the application; and the evaluation phase, where the completed product underwent rigorous testing, validation, and systematic assessment against established criteria and benchmarks.

During the evaluation stage, the study objectively measured the application's technical performance, operational reliability, security measures, and overall user acceptability through appropriate evaluation criteria, controlled testing environments, and structured feedback mechanisms.

By utilizing this design, the research aims not only to produce a fully functional and operational application but also to scientifically validate that it effectively meets all intended objectives, successfully solves the targeted problems, and adheres to established industry standards for quality, efficiency, and data integrity. This methodological approach ensures that every step of the development process is carefully documented, analyzed, and justified, leading to a final output that is both technically sound and practically useful in real-world educational settings.

Furthermore, the developmental research design allowed the researchers to engage in continuous improvement and refinement of the proposed system throughout the entire duration of the study. Feedback gathered from actual users, expert evaluators, and systematic testing procedures served as the primary basis for modifying features, correcting errors, enhancing interface design, and boosting overall system performance—for instance, an internal security review conducted after the initial prototype was completed identified several access-control gaps and functional defects, which were subsequently corrected before the system was finalized for user acceptance testing, as detailed in Section 3.6.

Through this iterative process, the researchers were able to identify possible issues, technical limitations, or gaps in functionality early in the cycle, allowing for necessary adjustments before the final implementation and deployment of the system. This ensures that

the developed application remains responsive to the actual, evolving needs of the intended users and is fully capable of delivering efficient, secure, and reliable services.

In addition, the study incorporated a mixed-method approach, utilizing both qualitative and quantitative methods in assessing the effectiveness, efficiency, and impact of the application. Qualitative data included direct observations, detailed user comments, and constructive suggestions regarding the usability, navigation, and functionality of the system, while quantitative data involved statistical analysis of survey results, numerical evaluation ratings, and objective measurements of system speed and accuracy.

The strategic combination of these methods provides a more comprehensive, holistic assessment of the developed application and strengthens the reliability, validity, and depth of the research findings. The chosen research design is also appropriate because it actively supports innovation and problem-solving through systematic development procedures, enabling the researchers to effectively apply theoretical knowledge, technical skills, and best practices in software engineering to produce a practical solution that directly addresses the identified problems and operational gaps.

Moreover, the developmental research design emphasizes continuous monitoring, verification, and detailed documentation throughout the entire process of system creation. Every stage of development—from initial planning and requirements analysis to coding, testing, and final validation—was carefully recorded to ensure full transparency, traceability, and consistency in the conduct of the study, serving as a valuable reference for future improvements, system maintenance, feature expansion, or further academic inquiry related to the project.

3.2 System Development Methodology

The Agile Development Model was used, consisting of the following phases:

Requirements Analysis - The study identified and documented the necessary features, functions, and constraints of the application by gathering data from potential users, existing literature, and related systems. This step defined the scope and established clear objectives to guide the development. In this phase, the researchers also determined the problems encountered by users and analyzed the current processes that needed improvement. Interviews, surveys, and observations were conducted to collect relevant information that served as the basis for designing the proposed system, ensuring that the application addresses the actual needs and expectations of the target users.

System Design - This phase involved translating the gathered requirements into a detailed technical blueprint, including the architecture, user interface layout, database structure, and system workflows to ensure efficiency and user-friendliness. This phase focused on creating an organized and visually appealing design that enhances the overall user experience. The researchers prepared wireframes, database diagrams, and use case diagrams to illustrate how the system would function. Proper planning during this stage helped minimize errors during development and ensured that all components of the system work together effectively.

Development - The actual coding, database construction, and integration of modules took place during this phase, turning the conceptual design into a working software component. Development was done in cycles, allowing parts of the system to be built and reviewed progressively. The iterative nature of Agile enabled continuous improvement and flexibility

whenever modifications were needed. The researchers utilized appropriate programming languages, frameworks, and development tools to ensure the quality and reliability of the application.

Testing - The developed features underwent various assessments such as functional testing, usability testing, security/access-control testing, and error checking to identify bugs, vulnerabilities, and areas for enhancement before finalization. This phase was essential to ensure that the application operates correctly and securely, and delivers accurate results. Different testing methods were applied to evaluate the reliability, compatibility, security, and responsiveness of the system. Feedback gathered from testers was analyzed carefully to identify necessary improvements, and detected issues were corrected immediately to enhance the overall performance, security, and usability of the application.

Deployment (Prototype) - This phase allowed for practical implementation and realworld evaluation of whether the system meets its intended purpose. By applying the Agile Development Model, the research ensured that the resulting application is not only technically robust and secure but also highly responsive to user needs. During this phase, the developed prototype was introduced to selected users for actual use and assessment, with their feedback and experiences providing valuable insights regarding the strengths and weaknesses of the system, leading to necessary revisions before final release.

Furthermore, the Agile Development Model encourages collaboration, adaptability, and continuous improvement throughout the development process. Unlike traditional models, Agile allows developers to respond quickly to changes and user feedback, making it highly suitable for projects that require flexibility and ongoing enhancement. Through regular communication,

progress evaluation, and iterative development cycles, the researchers were able to ensure that the system continuously evolves according to user needs and technological requirements.

3.3 System Architecture

The system follows a structured client-server architecture, a widely adopted and reliable model that effectively divides the overall system into two distinct yet interconnected components. In this configuration, end-users—including school administrators, teachers, and parents—interact with the system through accessible, responsive web-based interfaces, while a centralized server infrastructure serves as the processing core responsible for handling data operations, enforcing access-control and security rules, and managing the secure storage, organization, and retrieval of all institutional data.

This architectural model establishes a clear separation of responsibilities between the client side and the server side, ensuring organized functionality, optimized operational performance, and significantly easier maintenance, updates, and future scalability as user demand or institutional requirements grow over time.

The client side serves as the front-end layer, which is designed to be intuitive, visually clear, and user-friendly; it is primarily responsible for presenting information in a readable format, capturing user inputs and commands through forms or interactive elements, and displaying processed results or feedback retrieved from the server. This interface layer was developed using modern, responsive technologies and design principles to ensure compatibility across different devices, screen sizes, and operating systems, thereby providing a consistent, accessible user experience whether the system is accessed via a standard web browser on a desktop computer or a mobile device.

On the other hand, the server acts as the central backend component and computational engine of the entire system. It hosts the core application logic, enforces rolebased access control and other security protocols (see Section 3.4), processes form submissions, and manages internal communications and transactions. When a user submits a request through the interface, the request first passes through a centralized authentication and authorization layer that verifies the user is logged in and holds the correct role for the page being requested, before the server processes the input, executes the required business logic, and retrieves or stores relevant records within the database.

The database, hosted and managed entirely at the server level, serves as the secure, structured repository for all system data, including user accounts, student records, attendance logs, grades, inventory records, transaction logs, and audit trails, all organized and indexed using a normalized relational schema to ensure fast retrieval, long-term data integrity, and referential consistency.

By centralizing the processing power, business logic, and data storage resources on the server side, this architecture ensures that system updates, functional improvements, and security patches can be implemented once and deployed centrally without requiring modifications or installations on every individual user device, while also enabling multiple authorized users to access the system simultaneously, reliably, and without conflict. This structural design ultimately supports the development of a robust, secure, and highperforming system that meets the functional requirements, performance standards, and strategic goals outlined in the study.

3.4 Tools and Technologies Used

The development and deployment of the system utilized the following technologies, carefully selected and evaluated based on their accessibility, performance, ease of integration, security features, and overall suitability for the project's defined scope, functional requirements, and target operating environment, ensuring that every component works harmoniously to deliver a stable, efficient, and secure web-based solution.

Frontend: The user interface and client-side structure were built using HTML (HyperText Markup Language), which serves as the fundamental structural foundation for all web pages, combined with Tailwind CSS, a utility-first styling framework that enables the creation of modern, responsive, and consistent designs that render correctly across different screen sizes and devices, ranging from desktop computers to tablets and mobile phones. JavaScript is used throughout the interface to add interactivity and, specifically through AJAX (Asynchronous JavaScript and XML), to power the parent-teacher messaging feature's periodic-refresh (polling) behavior described in Section 3.4's Real-time Messaging discussion below, without requiring the browser to reload the entire page.

Backend: All server-side logic, data processing, and core operational functionalities are powered by PHP, a mature, widely-used open-source scripting language well-suited to web development. To prioritize data security and integrity, the system uses PDO (PHP Data Objects) with prepared statements for every single database interaction, a practice that neutralizes SQL injection — one of the most common and damaging vulnerabilities found in web applications — ensuring that all sensitive information, including student records and login credentials, remains protected from unauthorized access, interception, or manipulation.

Database: MySQL serves as the primary, centralized, and fully relational data storage solution for the entire platform, selected for its reliability, scalability, and compatibility with the chosen PHP development environment. It stores and manages all essential system data, including student profiles, enrollment records, attendance logs, grades, inventory records and transaction history, user account credentials, communication archives, and audit logs. The database was designed with a normalized schema to reduce redundancy, maintain referential integrity through foreign key constraints, and support efficient querying and reporting.

Application-Level Security: Beyond the database-level protections described above, the system implements the following application-layer security measures, developed and refined as part of the study's testing and revision process described in Section 3.6:

- Role-Based Access Control (RBAC): every protected page requires the user to be authenticated and holds a specific role (Administrator, Teacher, or Parent) before any content is rendered. This check happens on the server, centrally, before the page produces any output, closing a class of vulnerability known as "broken access control," where a user could otherwise reach another role's pages simply by navigating to the URL directly.
- Cross-Site Request Forgery (CSRF) protection: every form that changes data (adding, editing, or deleting a record) includes a unique, session-bound token that the server verifies before processing the request, preventing an attacker's website from silently submitting actions on behalf of a logged-in user.
- Password security: user passwords are never stored in plaintext. They are hashed using PHP's `password_hash()` function (bcrypt algorithm) and verified using `password_verify()`, meaning that even direct access to the database would not reveal a usable password.
- Session hardening: session cookies are configured as `HttpOnly` (inaccessible to clientside scripts) and `SameSite` (restricting cross-site use), sessions are regenerated on

login to prevent session fixation, and idle sessions automatically expire after 30 minutes.

- Login rate limiting: repeated failed login attempts for the same username are temporarily blocked after five failures within a fifteen-minute window, mitigating brute-force password guessing.
- Audit logging: significant actions (record creation, editing, deletion, and login events) are written to an audit log table capturing the acting user, the action, and a timestamp, supporting accountability and troubleshooting as described in Objective 1.3.1.
- File upload validation: uploaded profile images are validated both by file extension and by verifying the file is a genuine image (rejecting disguised non-image files), and uploaded files are stored in directories configured to prevent script execution.

Real-time Messaging: To facilitate timely communication within the Parent-Teacher Portal, the system uses AJAX polling, where the browser periodically checks the server (every few seconds) for new messages or announcements in the background. This approach was specifically selected because it achieves near real-time functionality without requiring the installation or maintenance of additional server software, specialized extensions, or paid infrastructure such as a dedicated WebSocket server, making it well-suited to the school's budgetary considerations and the technical limitations of typical shared web hosting environments.

Development Tools: The entire coding and development process was carried out using Visual Studio Code, a lightweight and widely used source-code editor offering syntax highlighting, debugging support, and a wide library of extensions that streamline writing and maintaining clean, structured code. Git was used for version control, allowing the development team to track revisions, collaborate on shared files, and safely revert to previous stable versions of the code if errors arose during development.

Deployment Platform (Demo Hosting): For online access, public demonstration, and formal evaluation purposes, the system is deployed on ProFreeHost, a free web hosting service that supports the PHP and MySQL technology stack required by the application. This platform provides a cPanel interface for management, supports recent PHP versions, includes MySQL database management tools, and offers a free subdomain for the demonstration deployment. As a free shared hosting service, this platform comes with certain limitations, such as constrained storage capacity and performance that may fluctuate depending on server load; these constraints are acceptable and sufficient for the purpose of presenting a functional prototype for capstone evaluation.

Internal School Access: In addition to the cloud-based online demo version, the system is also made accessible within the school's premises through the Local Area Network (LAN) using XAMPP, a popular open-source software package that provides a complete local server environment including Apache, PHP, and MySQL. This setup allows teachers, administrators, and staff physically located on campus to access and use the system with optimal speed and zero dependency on external internet connectivity. This dual deployment strategy — combining both online hosting and local network access — ensures that the system can be evaluated, demonstrated, and utilized in both online and offline/local settings, demonstrating its flexibility and readiness for potential full-scale implementation within the school's daily operations.

Overall, the selected technologies, frameworks, and development tools collectively form a stable, efficient, and security-conscious technical foundation that supports the operational, administrative, and academic needs of Tigbauan Fundamental Learning Center, Inc., while providing a smooth, accessible experience for all intended users, including administrators, teachers, parents, and students.

3.5 Data Gathering Techniques

To ensure that the resulting system fully aligns with the unique operational requirements, specific challenges, and long-term strategic goals of the institution, a comprehensive, multi-layered research methodology was designed and employed throughout the preliminary phase of the study.

This approach drew heavily from established qualitative and analytical research techniques, effectively combining the collection and examination of historical data patterns with direct feedback and perspectives from key stakeholders to form a complete and accurate picture of the school's environment. By utilizing this integrated strategy, the research team was able to gain not only a deep understanding of the school's current operational ecosystem and existing procedural limitations but also laid down a robust, evidence-based foundation for designing a highly customized, user-centric digital solution.

Document Analysis: A thorough, systematic review of existing physical and digital school records was conducted to ensure a complete and accurate understanding of historical practices, established standards, and recurring operational procedures currently in place at Tigbauan Fundamental Learning Center, Inc. The scope of this review covered a range of official documents, including handwritten attendance sheets and logbooks, manual inventory logs, enrollment and admission forms, quarterly and final grade reports, and administrative circulars.

This process involved cataloging, classifying, and organizing these documents, as well as digitizing samples where feasible, followed by content analysis and data mapping. Through this method, the team was able to identify and map out essential data fields and variables—such as unique student identifiers, timestamps, and item quantities—while also

detecting common redundancies, information gaps, and bottlenecks present in the current manual workflows.

Furthermore, this analysis allowed the researchers to outline the precise logic, sequence of operations, and functional dependencies required for a successful digital transition, establishing the standard data formats, validation rules, and reporting structures that the new system must support.

Semi-Structured Interviews: Complementing the document analysis, qualitative interviews were conducted using a semi-structured format, guided by open-ended questions but designed to remain flexible enough to allow for spontaneous probing and natural flow of discussion. These sessions were held individually and in small focus groups with a diverse cross-section of participants directly involved in the school's operations, including school administrators, senior faculty members, subject teachers, and select non-teaching support staff responsible for inventory and general maintenance.

Each interview session lasted approximately 45 to 60 minutes, and in total, the research team engaged with over 20 participants representing various departments and roles to ensure a balanced and comprehensive collection of insights.

The discussions focused extensively on identifying and documenting key pain points and inefficiencies inherent in the current manual and paper-based systems—such as timeconsuming data entry, frequent calculation errors, difficulties in locating physical documents, delays in report generation, and persistent communication gaps between the administration, teachers, and parents. Beyond identifying problems, these sessions were instrumental in gathering functional and non-functional requirements for the proposed digital

platform directly from the perspective of the end-users, including desired features such as mobile accessibility, clearly defined role-based access permissions, and automated alerts.

This deep understanding allowed the development team to clearly define distinct access levels and specialized system modules tailored to the varying needs and responsibilities of administrators, teachers, and parents, ensuring that each user only sees and interacts with the features relevant to their role.

3.6 Testing Procedures

To guarantee the quality, reliability, security, and performance of the Web-Based Integrated School Management System, a multi-layered, systematic testing methodology was implemented throughout the entire development lifecycle. This validation process followed a structured progression from individual code verification to full system evaluation, security review, and finally real-world user validation, ensuring that every aspect of the application met the predefined technical specifications, functional requirements, and operational standards established for Tigbauan Fundamental Learning Center, Inc.

Unit Testing: This phase represents the most granular level of software validation, focusing on the individual components, modules, and functions that form the building blocks of the system. During this stage, specific features such as the attendance-recording logic, grade computation and validation, inventory stock calculations, and database query handlers were tested in isolation from the rest of the application. Test cases covering standard operational inputs, boundary values, and invalid or unexpected data formats were used to verify the correctness and robustness of each unit, ensuring that it consistently produces the expected output and handles exceptions gracefully without compromising data integrity.

System Testing: Following the successful validation of individual units and the completion of system integration, a comprehensive end-to-end evaluation of the fully integrated system was conducted. The primary goal was to assess overall functionality, operational performance, and data integrity across interconnected modules — for instance, verifying that a grade entered by a teacher is correctly reflected on the parent's grade view, that attendance marked in the teacher module correctly appears in the corresponding parent and admin reports, and that inventory transactions are correctly logged when stock is added or adjusted. The scope of testing also included cross-browser compatibility checks to ensure consistent appearance and functionality across different browsers and devices.

Security and Access-Control Testing: In addition to functional testing, the system underwent a dedicated review focused specifically on access control and common web application vulnerabilities, guided by categories from the OWASP (Open Web Application Security Project) Top 10. This review involved manually attempting to access role-restricted pages (for example, a teacher-only page) while logged in under a different role, or while not logged in at all, to confirm that the server correctly denies access rather than relying solely on hiding the corresponding menu link in the interface. It also involved verifying that statechanging requests without a valid CSRF token are rejected, that uploaded files are validated as genuine images before being accepted, and that database queries are not vulnerable to SQL injection due to the consistent use of prepared statements. Issues identified during this review — including several pages that lacked a server-side role check and two internal developer utility scripts that lacked any authentication — were corrected and re-tested before the system was considered ready for User Acceptance Testing.

User Acceptance Testing (UAT): Serving as the final validation phase, User Acceptance Testing involved a collaborative evaluation process conducted with the direct participation of actual end-users representing the school community, including school administrators, teaching faculty, non-teaching staff, and selected parents. Participants were tasked with performing realistic operational activities relevant to their roles — such as processing new student admissions, marking daily attendance, encoding grades, monitoring inventory levels, sending communication messages, and generating reports — as they would do in regular operations.

Throughout the testing period, structured feedback was gathered through usability observations and debrief discussions, where participants shared their opinions regarding the system's usability, ease of navigation, and overall alignment with established school workflows. All feedback was compiled, categorized, and prioritized, leading to targeted refinements in interface design, workflow logic, and system functionality, with multiple rounds of testing and revision conducted until participants expressed confidence and satisfaction with the application.

Collectively, these layered testing procedures — spanning Unit Testing, System Testing, Security and Access-Control Testing, and User Acceptance Testing — provided a complete evaluation framework that ensured the reliability, quality, security, and usability of the developed application before its deployment.

3.7 Evaluation Criteria

The system was evaluated against the following target benchmarks:

Metric	Target Benchmark
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Response Time	High-priority interactions — saving attendance records, updating inventory quantities, loading student profiles, and generating reports — are acknowledged and reflected in the interface within 2 seconds under normal load, so that navigation and data entry feel immediate rather than sluggish.
Concurrent User Support	The system remains stable while supporting at least 50 simultaneous users performing various tasks, simulating real-world conditions such as the start of classes or an inventory audit, without slowdowns, crashes, or data conflicts.
System Uptime	The platform remains accessible during school hours with a target reliability of approximately 99.9% uptime, so that administrative and academic processes are not interrupted by unexpected downtime.
Access Control Accuracy	Every attempt to access a page outside a user’s assigned role (e.g., a parent account requesting a teacher-only page) is correctly denied and redirected, verified through the security testing described in Section 3.6.

3.8 System Design

3.8.1 Data Flow Diagram (Level 0)

The Level 0 Data Flow Diagram (also known as a Context Diagram) presents the highest-level abstraction of the system, illustrating its overall scope, its external users, and the primary flow of information between those users, the system's core processes, and the central database. It simplifies the complex internal operations of the Web-Based Integrated School Management System into a clear, easy-to-understand overview.

The diagram is structured around four external entities that interact with the system — the Administrator, Teacher, Parent, and the central Database — surrounding a single process representing the entire School Management System.

The data flow follows a logical sequence of input, processing, storage, and output:

- The Administrator submits student records, user account details, subject/inventory data, and receives consolidated reports and dashboards in return.

- The Teacher submits attendance records and grade entries, and receives class rosters and student information in return.
- The Parent submits messages to teachers/admin, and receives attendance records, grades, and announcements in return.
- The System Process validates and processes all of the above according to each user's role and permissions, and reads from / writes to the Database, which stores all student, attendance, grade, inventory, user, message, and audit data.
- Processed Results (reports, dashboards, confirmations) are returned from the System Process back to the requesting user.

This diagram effectively captures the fundamental interaction cycle, providing a visual foundation for understanding how information is captured, transformed, retained, and utilized within the system, and serves as a reference point for the more detailed Entity Relationship Diagram and Use Case Diagram presented in the following sections.

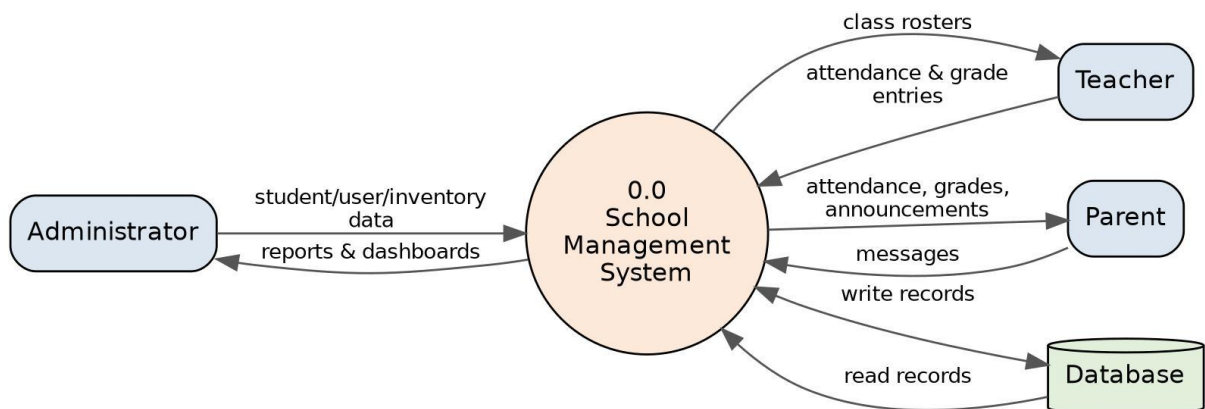


Figure 3.1 — Level 0 Data Flow Diagram (Context Diagram) of the Web-Based Integrated School Management System

3.8.2 Entity Relationship Diagram (ERD)

The Entity Relationship Diagram (ERD) serves as the conceptual blueprint of the database architecture, illustrating the structure, essential data attributes, data types, and logical associations that govern how information is captured, stored, related, and interconnected within the proposed school management system.

By modeling these connections — whether one-to-one, one-to-many, or many-to-many — the ERD ensures data integrity and consistency throughout the system, minimizing unnecessary data redundancy and establishing referential integrity rules that prevent orphaned or conflicting records. It also forms the basis for the physical database schema, defining tables, fields, primary keys, and foreign keys.

Below is the breakdown of each core entity, its key attributes, and its relationships within the database structure:

Users (id, username, password, role, name, email, phone, profile_image, created_at) — This entity is the central authentication table for every person authorized to log into the system: administrators, teachers, and parents. The role column, restricted to admin, teacher, or parent, drives the Role-Based Access Control described in Section 3.4 — it determines exactly which pages and functions a given account may access. Passwords are stored only as bcrypt hashes, never in plaintext.

Students (student_id, name, grade_level, gender, dob, address, admission_date, status, parent_phone, profile_image) — This entity stores each learner's core profile and enrollment information. student_id is the primary key referenced by every other module that concerns a specific learner (attendance, grades, and the student-parent relationship).

Student_Parent (student_id, parent_id, relation) — A linking (junction) table that implements the many-to-many relationship between Students and Users, allowing a parent account to be associated with one or more children, and, if needed in the future, a student to be associated with more than one guardian account.

Attendance (id, student_id, date, status, notes, created_at) — Records each day's attendance status (Present, Absent, Late, Excused) for a given student, entered by the class teacher. status and date together support the attendance-percentage and trend reports described in Objective 1.3.2 (#1).

Subjects (id, subject_name, grade_level) — Stores the list of subjects offered per grade level, maintained by the Administrator, and referenced by the Grades entity below.

Grades (id, student_id, subject_id, teacher_id, school_year, quarter, grade, remarks) — Records a student's quarterly grade for a specific subject and school year, entered by the subject teacher. The combination of student_id, subject_id, school_year, and quarter is constrained to be unique, so that re-encoding a grade updates the existing record rather than creating a duplicate.

Inventory (id, item_name, category, quantity, min_stock, borrowed, location, last_updated) — Stores the current state of each school asset or supply item, including its current quantity and the minimum stock threshold used to trigger a low-stock indicator.

Inventory_Transactions (id, item_id, action, quantity, performed_by, notes, created_at) — Logs every change made to an inventory item's stock (Restock, Adjustment, Disposal) along with who made the change and when, providing the transaction history described in Objective 1.3.2 (#4).

Messages (id, sender_id, receiver_id, message, created_at, read_status) — Stores each message sent between a parent, teacher, or administrator, supporting the Parent-Teacher Portal described in Objective 1.3.2 (#5).

Audit_Log (id, user_id, action, details, ip_address, created_at) — Records significant system actions (record creation, editing, deletion, and login events) for accountability and traceability, as described in Section 1.3.1 and Section 3.4.

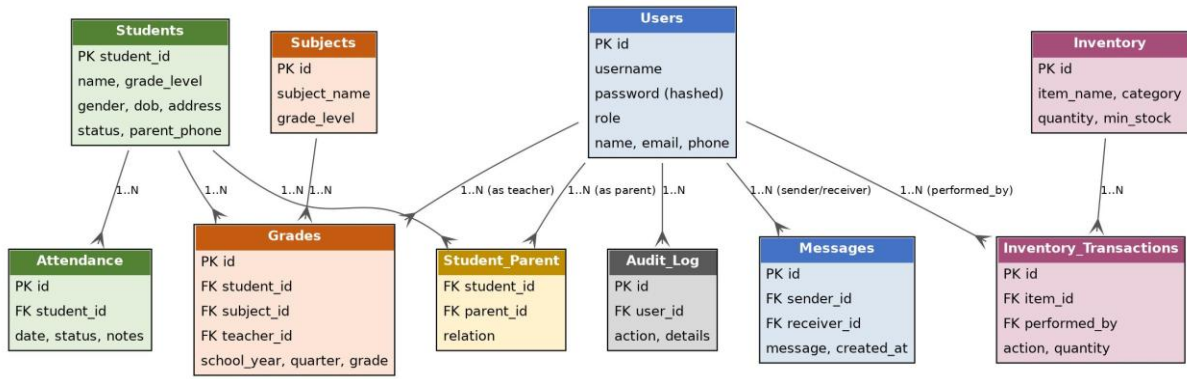


Figure 3.2 — Entity Relationship Diagram of the Web-Based Integrated School Management System

3.8.3 Use Case Diagram

The Use Case Diagram serves as a visual representation that comprehensively illustrates the functional requirements, operational scope, and intended behavior of the Integrated School Management System by defining and mapping out the significant interactions that occur between the platform and its various users.

This diagram identifies the three actors involved — Administrator, Teacher, and Parent — along with their specific roles and responsibilities, and precisely maps out the range of functionalities accessible to each distinct user type. It effectively categorizes features strictly based on user role, visually reinforcing the principle that access rights must align precisely with assigned institutional responsibilities, as enforced by the Role-Based Access Control described in Section 3.4.

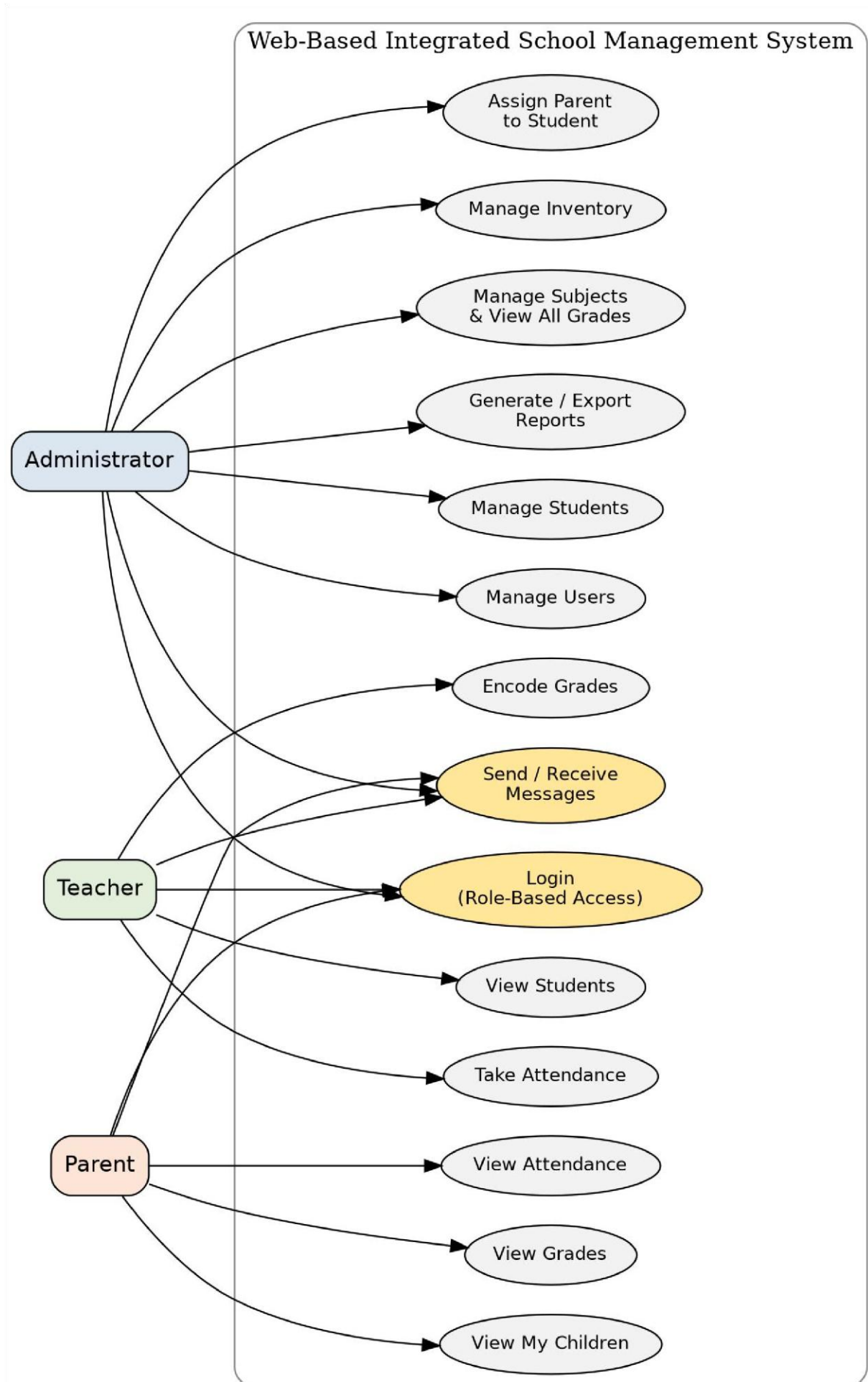


Figure 3.3 — Use Case Diagram of the Web-Based Integrated School Management System

Below is the breakdown of actors and their respective use cases:

Administrator Use Cases:

- Manage Students: Add, edit, view, and remove student records; upload student photos.
- Manage Users: Create, edit, and remove teacher and parent accounts; assign roles.
- Assign Parent to Student: Link a parent account to one or more student records.
- Manage Inventory: Add new inventory items, adjust stock levels, and review the inventory transaction log.
- Manage Subjects & Grades: Maintain the list of subjects per grade level and review all grades recorded across the institution.
- Generate Reports: View and export (CSV/Excel) student masterlists, attendance summaries, grade reports, and inventory reports.
- View Audit Log: Review the log of significant system actions for accountability purposes.

Teacher Use Cases:

- Take Attendance: Mark daily attendance (Present, Absent, Late, Excused) for students in their assigned grade level.
- Encode Grades: Enter and update quarterly grades for students per subject.
- View Students: View the list and profiles of students in their assigned classes.
- Send Messages: Communicate with parents or the administrator through the messaging module.

Parent Use Cases:

- View My Children: View the profile(s) of the student(s) linked to their account.

- View Attendance: View their child's attendance history and computed attendance percentage.
- View Grades: View their child's recorded grades per subject and quarter, and quarterly averages.
- Send Messages: Communicate with their child's teacher(s) or the administrator through the messaging module.

Register Account & Login (shared across roles): Every user account is created by the Administrator through the Manage Users function rather than via open self-registration, consistent with the institution's need to strictly control who is issued access to student data. Once an account exists, the user logs in with a username and password; the system verifies the password hash, applies the session-security measures described in Section 3.4, and directs the user to the dashboard corresponding to their role.

3.9 Requirements Specification

This section consolidates the functional and non-functional requirements of the proposed system, derived from the data-gathering procedures described in Section 3.5 and from the use-case analysis presented in Section 3.8. These requirements served as the basis against which the developed prototype was validated during testing (Section 3.6) and evaluation (Section 3.7).

3.9.1 Functional Requirements

- The system shall allow the Administrator to create, edit, view, and deactivate student records, including uploading a student photo.
- The system shall allow the Administrator to create, edit, and remove teacher and parent user accounts, and to assign each account its corresponding role.

- The system shall allow the Administrator to link one or more parent accounts to a student record.
- The system shall allow Teachers to record daily attendance (Present, Absent, Late, or Excused) for the students in their assigned grade level or section.
- The system shall allow Teachers to encode and update quarterly grades per subject for the students in their assigned classes.
- The system shall automatically compute quarterly and annual grade averages based on encoded subject grades.
- The system shall automatically compute a student's attendance percentage based on recorded daily attendance.
- The system shall allow the Administrator to add inventory items, adjust stock levels, and review a chronological log of inventory transactions.
- The system shall notify the Administrator when an inventory item's stock falls below a configurable reorder level.
- The system shall allow Parents to view the profile, attendance history, and recorded grades of the student(s) linked to their account.
- The system shall provide a messaging module through which Teachers, Parents, and the Administrator can exchange messages related to a specific student.
- The system shall allow the Administrator to generate and export (in CSV or Excel format) student masterlists, attendance summaries, grade reports, and inventory reports.
- The system shall maintain an audit log recording significant actions performed by users, including the actor, the action taken, and the timestamp.
- The system shall restrict every function listed above according to the authenticated user's assigned role, denying access to functions outside that role's permissions.

3.9.2 Non-Functional Requirements

Category	Requirement
Performance	Common pages(dashboard, attendance sheet, grade sheet) shall load within

	approximately three seconds under normal school-network conditions.
Security	Passwords shall be stored using a one-way cryptographic hash; all administrative and data-entry actions shall require an authenticated session; role-based access control shall be enforced on every request.
Usability	A user with basic computer literacy shall be able to complete core tasks (e.g., taking attendance, encoding a grade) after no more than one short orientation session, consistent with the ease-of-use concern raised in Section 1.7.
Reliability	The system shall correctly compute attendance percentages and grade averages consistent with the institution's existing grading policy, with no data loss during normal operation.
Maintainability	The system's codebase shall be organized into clearly separated modules (student records, attendance, grades, inventory, messaging, reporting) to simplify future maintenance and feature additions.
Scalability	The underlying database design shall accommodate growth in enrollment and historical records across multiple school years without requiring a redesign of the schema.
Compatibility	The system shall be accessible through standard, commonly available web browsers on both desktop and mobile devices, without requiring the installation of specialized software.

Table 3.1 — Non-Functional Requirements of the Proposed System

3.10 Data Dictionary (Overview)

The complete data dictionary, listing every field, data type, and description for each of the system's core database entities, is provided in Appendix F. As an overview, Table 3.2 below presents the Students entity, which serves as the central record to which attendance, grades, and parent-linkage data are related.

Field Name	Data Type	Description
student_id	INT (PK)	Unique system-generated identifier for the student record.
lrm	VARCHAR(20)	Learner Reference Number issued by the Department of Education, where available.
first_name / last_name	VARCHAR(100)	Given name and surname of the student.
birthdate	DATE	Student's date of birth, used for age verification and reporting.
grade_level	VARCHAR(20)	Current grade level of the student (e.g., Grade 4).
section	VARCHAR(50)	Section or class grouping to which the student is currently assigned.
photo_path	VARCHAR(255)	File path to the student's uploaded photo, if any.
status	ENUM	Enrollment status of the record (Active, Inactive, Transferred, Graduated).
date_created	DATETIME	Timestamp indicating when the record was created in the system.

Table 3.2 — Sample Data Dictionary Entry: Students Table

3.11 Feasibility Analysis

3.11.1 Technical Feasibility

The proposed system is technically feasible because it relies exclusively on widely available, well-documented, and predominantly free or low-cost web technologies, as listed in Section 3.4. The institution already has basic internet connectivity, and the system is designed to run on commodity, low-cost web hosting, avoiding the need for specialized servers or proprietary infrastructure. The researchers' existing familiarity with the chosen technology stack, developed through prior coursework, further supports the technical feasibility of completing the system within the capstone timeline.

3.11.2 Operational Feasibility

The system is operationally feasible because it is designed to mirror the institution's existing workflows — enrollment, daily attendance, quarterly grading, inventory tracking, and parent communication — rather than imposing an entirely new process on its users. Because the Administrator and Teachers are already performing these tasks manually, the system automates and streamlines familiar work rather than introducing unfamiliar responsibilities, which reduces the resistance to adoption anticipated by the Diffusion of Innovation Theory discussed in Section 1.7.

3.11.3 Economic Feasibility

From a cost perspective, the proposed system is expected to be substantially less expensive to operate over time than the continued use of paper-based processes, once the costs of paper, printing, physical storage, and staff time spent on manual retrieval and computation are considered. Table 3.3 presents an indicative, order-of-magnitude comparison of anticipated

costs and benefits; precise figures will depend on the final hosting arrangement and are expected to be refined during Capstone 2.

Cost / Benefit Item	Estimated Impact
Web hosting and domain (annual)	Low, recurring cost; typically a small fraction of current annual paper and printing expenses.
Developer time (capstone researchers)	Absorbed as part of the academic requirement; no direct monetary cost to the institution.
Staff training / orientation	One-time, low cost in staff hours, mitigated by the usability requirements in Table 3.1.
Reduced paper, printing, and physical storage costs	Recurring savings, realized progressively as records are digitized.
Reduced staff time on manual retrieval and computation	Recurring efficiency gain, freeing staff time for instructional and administrative priorities.

Table 3.3 — Indicative Cost-Benefit Comparison

3.11.4 Schedule Feasibility

The development schedule, detailed in Section 3.12 and Appendix G, allocates sufficient time within the Capstone 1 and Capstone 2 timeframe for requirements analysis, design, iterative development, testing, and documentation. The phased Agile approach described in Section 3.2 allows the researchers to adjust the scope of later iterations if unforeseen delays occur in earlier phases, without jeopardizing the delivery of the system's core functions.

3.12 Project Development Timeline

Table 3.4 summarizes the planned development timeline across the two capstone terms. A detailed, week-by-week breakdown is provided in Appendix G.

Phase	Approximate Duration	Key Deliverable
Requirements Analysis & Planning	Weeks 1–3	Approved capstone proposal; finalized requirements specification.
System Design	Weeks 4–6	Data flow diagrams, entity relationship diagram, use case diagram, database schema.
Development – Core Modules	Weeks 7–11	Working modules for student records, attendance, and grades.

Development – Supporting Modules	Weeks 12–14	Working modules for inventory and messaging; reporting/export functions.
Testing	Weeks 15–16	Functional, usability, and security/accesscontrol test results.
Evaluation & Documentation	Weeks 17–18	User acceptance evaluation results; final documentation and manuscript.

Table 3.4 — Summary Project Development Timeline

3.13 Ethical Considerations

Because the proposed system stores personal information belonging to minors, the study observes several ethical safeguards throughout its data-gathering and development activities. Prior to conducting interviews and surveys with the school administrator, teachers, and parents, the researchers secured informed consent from each participant, explaining the purpose of the study, the voluntary nature of participation, and the manner in which the gathered information would be used solely for the purposes of this capstone project (see Appendix B).

Any sample student data used during development and testing is anonymized or fictionalized rather than drawn from actual learner records, consistent with the data minimization principle underlying the Data Privacy Act discussed in Section 2.5. Access to any real institutional data made available by the school for reference purposes is limited to the researchers directly involved in the study and is not reproduced in this document beyond what is necessary to illustrate system requirements.

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APPENDICES

The following appendices provide supporting documentation referenced throughout Chapters 1 to 3: request and consent instruments used during data gathering, the survey and interview instruments themselves, an illustration of the institution's current manual forms, the complete data dictionary, the detailed project timeline, blank curriculum vitae templates for the proponents, and a consolidated list of figures and tables.

Appendix A

Letter of Request to Conduct the Study

Date: _____

The School Director

Tigbauan Fundamental Learning Center, Inc.

Tigbauan, Iloilo

Dear Sir/Madam:

We are fourth-year Bachelor of Science in Information Technology students of PHINMA University of Iloilo, currently undertaking our Capstone Project titled “A Web-Based Integrated School Management System for Tigbauan Fundamental Learning Center, Inc.” In connection with this undertaking, we respectfully request permission to conduct interviews and administer survey questionnaires among your administrative staff, teachers, and parents, and to review de-identified samples of the institution’s current manual forms and processes, for the sole purpose of informing the design of the proposed system.

We assure you that all information gathered will be treated with strict confidentiality and used exclusively for academic purposes related to this capstone project, in accordance with the provisions of Republic Act No. 10173 (Data Privacy Act of 2012).

We would greatly appreciate your favorable consideration of this request. Thank you very much.

Respectfully yours,

Elgie Piosca Delos Reyes, Carl John M. Cero, Kylen Marie Sta. Maria, Eunice Melan
Lagunday

BSIT 3, PHINMA University of Iloilo

Noted by: _____ (Capstone Adviser)

Approved by: _____ (School Director)

Appendix B

Informed Consent Form

You are being invited to participate in a data-gathering activity for a capstone research project titled “A Web-Based Integrated School Management System for Tigbauan Fundamental Learning Center, Inc.,” conducted by BSIT students of PHINMA University of Iloilo. Your participation may involve an interview, a survey questionnaire, or both, and is expected to take no more than thirty minutes.

Your participation is entirely voluntary. You may decline to answer any question or withdraw at any point without any consequence. All responses will be treated confidentially, reported only in aggregate or anonymized form, and used solely for the purpose of informing the design of the proposed system.

By signing below, you confirm that you have read and understood the above, and that you voluntarily agree to participate.

Name (optional): _____

Role: ☐ Administrator ☐ Teacher ☐ Parent

Signature: _____ Date: _____

C

Survey Questionnaire on Current Practices and System Requirements

Instructions: Please rate the following statements using the scale: 4 – Strongly Agree, 3 – Agree, 2 – Disagree, 1 – Strongly Disagree.

Part I. Current Manual Process (for Administrators and Teachers)

- 1. Retrieving a specific student's record currently takes considerable time.
- 2. Errors occasionally occur in manually computed grades or attendance percentages.
- 3. Important student records are sometimes difficult to locate when needed.
- 4. Preparing periodic reports (e.g., masterlists, attendance summaries) manually is time-consuming.
- 5. Tracking school supplies and materials using the current logbook system is difficult.
- 6. Communicating with parents about a student's attendance or performance is currently inconvenient.

Part II. Anticipated System Requirements (for Administrators and Teachers)

- 1. A system that allows quick search and retrieval of student records would be valuable.
- 2. Automatic computation of attendance percentages and grade averages would reduce errors.
- 3. The ability to generate exportable reports (CSV/Excel) would save preparation time.
- 4. Role-based access (Administrator, Teacher, Parent) is important for data security.
- 5. An in-system messaging feature between teachers and parents would be useful.
- 6. Notifications for low inventory stock would help prevent shortages of school supplies.

Part III. Parent Portal Expectations (for Parents)

- 1. I would find it useful to view my child's attendance record online.

Appendix

- 2. I would find it useful to view my child's grades online, per quarter.
- 3. I would use a messaging feature to communicate with my child's teacher.
- 4. I am comfortable using a web-based system on a computer or mobile phone.
- 5. I would trust a school system with my child's personal information if access were properly restricted.

D**Interview Guide (Administrator)**

- 1. Can you walk us through how a new student is currently enrolled, from initial inquiry to the creation of a permanent record?
- 2. What are the most time-consuming or error-prone parts of the current attendance and grading process?
- 3. How is the school's inventory of supplies and materials currently tracked, and how are shortages typically discovered?
- 4. How does the school currently communicate attendance or performance concerns to parents?
- 5. What concerns, if any, would you have about moving these records to a web-based system?
- 6. What features would you consider most essential in a new system, and which would you consider optional?

Appendix E

Illustrative Layout of Current Manual Forms

The following tables recreate, for reference purposes, the general layout of the manual forms and logs currently used by the institution, as described by the Administrator during the interview process. They are reproduced here in simplified, blank form to illustrate the fields the proposed system must capture; no actual student data is shown.

E.1 Manual Daily Attendance Sheet (Layout)

No.	Student Name	Present	Absent	Late	Excused	Remarks

E.2 Manual Enrollment Form (Layout)

Field	Entry
Student Name	
Birthdate	
Grade Level / Section	
Parent / Guardian Name	
Contact Number	
Address	

E.3 Manual Inventory Logbook (Layout)

Date	Item	Qty In	Qty Out	Balance	Remarks

F

Complete Data Dictionary

F.1 Users Table

Field Name	Data Type	Description
user_id	INT (PK)	Unique identifier for the user account.
username	VARCHAR(50)	Login username, unique per account.
password_hash	VARCHAR(255)	One-way cryptographic hash of the user's password.
role	ENUM	Administrator, Teacher, or Parent.
full_name	VARCHAR(150)	Full name of the user.
contact_number	VARCHAR(20)	Contact number of the user, where provided.
status	ENUM	Active or Inactive account status.
date_created	DATETIME	Timestamp of account creation.

Field Name	Data Type	Description
grade_id	INT (PK)	Unique identifier for the grade record.

Appendix

student_id	INT (FK)	References the Students table.
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F.2 Parent-Student Link Table

Field Name	Data Type	Description
link_id	INT (PK)	Unique identifier for the parent-student link record.
parent_user_id	INT (FK)	References the Users table; identifies the parent account.
student_id	INT (FK)	References the Students table; identifies the linked student.
relationship	VARCHAR(30)	Relationship of the parent/guardian to the student.

F.3 Attendance Table

Field Name	Data Type	Description
attendance_id	INT (PK)	Unique identifier for the attendance record.
student_id	INT (FK)	References the Students table.
date	DATE	Date of the attendance record.
status	ENUM	Present, Absent, Late, or Excused.
recorded_by	INT (FK)	References the Users table; identifies the recording teacher.
remarks	VARCHAR(255)	Optional remarks (e.g., reason for absence).

F.4 Subjects Table

Field Name	Data Type	Description
subject_id	INT (PK)	Unique identifier for the subject.
subject_name	VARCHAR(100)	Name of the subject (e.g., Mathematics).
grade_level	VARCHAR(20)	Grade level to which the subject applies.

F.5 Grades Table

subject_id	INT (FK)	References the Subjects table.
quarter	ENUM	First, Second, Third, or Fourth Quarter.
grade_value	DECIMAL(5,2)	Numeric grade recorded for the quarter.
encoded_by	INT (FK)	References the Users table; identifies the encoding teacher.

F.6 Inventory Table

Field Name	Data Type	Description
item_id	INT (PK)	Unique identifier for the inventory item.
item_name	VARCHAR(150)	Name/description of the item.
quantity_on_hand	INT	Current recorded stock level.
reorder_level	INT	Threshold below which a low-stock notification is triggered.
unit	VARCHAR(20)	Unit of measure (e.g., pcs, box, ream).

F.7 Inventory Transactions Table

Field Name	Data Type	Description
transaction_id	INT (PK)	Unique identifier for the inventory transaction.
item_id	INT (FK)	References the Inventory table.
transaction_type	ENUM	Stock In or Stock Out.
quantity	INT	Quantity added or removed in this transaction.
date	DATETIME	Date and time of the transaction.

recorded_by	INT (FK)	References the Users table.
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F.8 Messages Table

Field Name	Data Type	Description
message_id	INT (PK)	Unique identifier for the message.
sender_id	INT (FK)	References the Users table; identifies the sender.
recipient_id	INT (FK)	References the Users table; identifies the recipient.
student_id	INT (FK)	References the Students table; associates the message with a student.
message_body	TEXT	Content of the message.
date_sent	DATETIME	Timestamp the message was sent.

F.9 Audit Log Table

Field Name	Data Type	Description
log_id	INT (PK)	Unique identifier for the audit log entry.
user_id	INT (FK)	References the Users table; identifies the actor.
action	VARCHAR(255)	Description of the action performed.
timestamp	DATETIME	Date and time the action occurred.

G

Detailed Project Timeline (Gantt Chart)

Week	Planned Activity
1–2	Problem identification, initial interview with Administrator
3	Survey administration to teachers and parents; data consolidation
4	Drafting of Chapter 1 (Introduction) and objectives
5	Review of related literature and studies (Chapter 2)
6	Synthesis of literature; legal and regulatory review
7	Requirements specification and use case analysis
8	Data flow diagram and entity relationship diagram design
9	Database schema creation
10–11	Development of student records and attendance modules
12	Development of grades module
13	Development of inventory module
14	Development of messaging and reporting modules
15	Functional and security/access-control testing
16	Usability testing and revisions
17	User acceptance evaluation
18	Finalization of documentation and manuscript

Appendix Appendix H

Curriculum Vitae (To Be Completed by Each Proponent)

Elgie Piosca Delos Reyes — BSIT 3

Section	Details
Personal Information	Address, contact number, and email address (to be completed by the proponent).
Educational Background	Elementary, secondary, and tertiary schools attended, with inclusive dates.
Skills	Relevant technical and soft skills (e.g., programming languages, tools, communication).
Seminars / Trainings Attended	Relevant seminars, workshops, or trainings related to information technology.
Character References	Name, position, and contact information of at least two character references.

Carl John M. Cero — BSIT 3

Section	Details
Personal Information	Address, contact number, and email address (to be completed by the proponent).
Educational Background	Elementary, secondary, and tertiary schools attended, with inclusive dates.
Skills	Relevant technical and soft skills (e.g., programming languages, tools, communication).
Seminars / Trainings Attended	Relevant seminars, workshops, or trainings related to information technology.
Character References	Name, position, and contact information of at least two character references.

Kylen Marie Sta. Maria — BSIT 3

Section	Details
Personal Information	Address, contact number, and email address (to be completed by the proponent).
Educational Background	Elementary, secondary, and tertiary schools attended, with inclusive dates.
Skills	Relevant technical and soft skills (e.g., programming languages, tools, communication).
Seminars / Trainings Attended	Relevant seminars, workshops, or trainings related to information technology.
Character References	Name, position, and contact information of at least two character references.

Eunice Melan Lagunday — BSIT 3

Section	Details
Personal Information	Address, contact number, and email address (to be completed by the proponent).
Educational Background	Elementary, secondary, and tertiary schools attended, with inclusive dates.

Skills	Relevant technical and soft skills (e.g., programming languages, tools, communication).
Seminars / Trainings Attended	Relevant seminars, workshops, or trainings related to information technology.
Character References	Name, position, and contact information of at least two character references.

Appendix I

List of Figures and Tables

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- Table 1.1 — Input-Process-Output Framework of the Study
- Table 3.1 — Non-Functional Requirements of the Proposed System
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- Table 3.4 — Summary Project Development Timeline
- Appendix F.1–F.9 — Complete Data Dictionary Tables
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Appendix J

Supplementary Review of Additional Related Literature and Studies

This appendix presents additional literature and studies consulted by the researchers that further support, but were not central enough to be included in, the main discussion in Chapter 2. They are grouped here by the same four thematic areas used in Sections 2.1 through 2.4.

J.1 On School Management and Student Information Systems

Beyond the works cited in Section 2.1, the researchers also reviewed studies examining the change-management dimension of adopting a school information system, which consistently note that successful adoption depends less on the sophistication of the software than on the clarity of the transition plan communicated to staff (Cuyegkeng, 2026). This reinforces the researchers' decision to pair the technical rollout described in Section 3.2 with a simple, phased orientation plan for the Administrator and Teachers, rather than a single, disruptive cutover.

Other studies on cloud-based student information systems emphasize that even small institutions benefit from centralizing student data in a single database rather than maintaining separate spreadsheets per class or subject, since centralization reduces the risk of conflicting versions of the same record (Al-Mutairi et al., 2023). This principle underlies the single, shared Students table described in the data dictionary in Appendix F, from which attendance, grades, and messaging records are all derived.

J.2 On Attendance Monitoring Systems

Additional literature on attendance systems highlights that the choice of identification method should be proportionate to the size and risk profile of the institution: large universities may justify the cost of biometric scanners, while small schools are generally better served by simpler, teacher-mediated digital attendance encoding (Garcia & Santos, 2023). This finding is consistent with the design decision, discussed in Section 2.6, to favor a teacher-encoded attendance module over biometric alternatives.

Other work on QR-code based attendance further notes that while such methods reduce encoding time, they introduce a dependency on students consistently carrying a physical or digital identifier, which can be impractical for younger learners (Abirami & Rohann, 2026).

This consideration further supports the researchers' preference for teacher-encoded attendance in the context of a K-12 institution serving young learners.

J.3 On Inventory and Asset Management

Supplementary literature on educational inventory systems observes that the majority of stock-related losses in small institutions stem not from theft but from simple undercounting and delayed reordering, both of which are effectively addressed by even a basic digital reorder-level alert, without requiring predictive analytics (Baliuag University, n.d.). This supports the rule-based reorder-level approach adopted in Section 3.11.1 in place of more complex forecasting models.

J.4 On Parent-Teacher Communication

Additional studies on parental engagement note that the frequency and specificity of communication — for example, a notification tied to a specific graded activity rather than a generic monthly newsletter — has a stronger association with parental follow-through than the particular technology platform used to deliver it (MPRA, 2025; SAJHE, 2025). This finding supports the decision to tie the messaging module directly to individual student records, described in Section 3.9.1, rather than implementing only a general broadcast announcement feature.

Appendix K

System User's Manual (Overview)

This appendix provides a preliminary, step-by-step guide to the core functions of the proposed system for each user role. It is intended to be refined and expanded with actual screen captures during Capstone 2, once the prototype is finalized.

K.1 Administrator Procedures

K.1.1 Logging In

- 1. Open the system URL in a web browser.
- 2. Enter the assigned username and password.
- 3. Click “Log In” to access the Administrator dashboard.

K.1.2 Adding a Student Record

- 1. From the dashboard, select “Manage Students.”
- 2. Click “Add New Student.”
- 3. Fill in the required fields (name, birthdate, grade level, section).
- 4. Optionally upload a student photo.
- 5. Click “Save” to create the record.

K.1.3 Creating a Teacher or Parent Account

- 1. Select “Manage Users.”
- 2. Click “Add New User.”
- 3. Enter the user's name and assign the appropriate role.
- 4. Click “Save”; the system generates the account and initial credentials.

K.1.4 Linking a Parent to a Student

- 1. Select “Manage Students” and open the student's record.
- 2. Click “Link Parent Account.”

- 3. Search for and select the parent's account.
- 4. Confirm the relationship and save.

K.1.5 Managing Inventory

- 1. Select "Manage Inventory."
- 2. Click "Add Item" to register a new supply item, or select an existing item to adjust stock.
- 3. Enter the quantity added or removed and the reason.
- 4. Save to update the stock level and transaction log.

K.1.6 Generating Reports

- 1. Select "Generate Reports."
- 2. Choose the report type (masterlist, attendance summary, grade report, or inventory report).
- 3. Select the relevant date range or grading period.
- 4. Click "Export" to download the report in CSV or Excel format.

K.2 Teacher Procedures

K.2.1 Taking Attendance

- 1. Log in and select "Attendance."
- 2. Select the class/section and date.
- 3. Mark each student as Present, Absent, Late, or Excused.
- 4. Click "Save Attendance."

K.2.2 Encoding Grades

- 1. Select "Grades" and choose the subject and quarter.
- 2. Enter each student's grade in the corresponding field.
- 3. Click "Save Grades"; the system automatically recomputes the quarterly average.

K.2.3 Messaging a Parent

- 1. Open a student's profile and select "Send Message."
- 2. Type the message and click "Send."
- 3. The message appears in the parent's message list.

K.3 Parent Procedures

K.3.1 Viewing a Child's Attendance

- 1. Log in to the Parent dashboard.
- 2. Select the linked child (if more than one).
- 3. Select "Attendance" to view the attendance history and computed percentage.

K.3.2 Viewing Grades

- 1. Select "Grades" from the Parent dashboard.
- 2. Choose the desired quarter to view recorded subject grades and the quarterly average.

K.3.3 Messaging a Teacher

- 1. Open the "Messages" tab.
- 2. Select the relevant teacher and type a message.
- 3. Click "Send."

Appendix L

Test Case Documentation

Table L.1 presents a representative subset of the functional test cases used during the testing phase described in Section 3.6. The complete set of test cases will be finalized and executed in full during Capstone 2.

Test ID	Module	Test Description	Expected Result
TC-01	Login	Log in with valid Administrator credentials.	User is redirected to the Administrator dashboard.
TC-02	Login	Log in with an incorrect password.	System displays an invalid-credentials message and denies access.
TC-03	Student Records	Add a new student record with all required fields completed.	Record is saved and appears in the student list.
TC-04	Student Records	Attempt to add a student record with a required field left blank.	System displays a validation error and does not save the record.
TC-05	Attendance	Teacher marks attendance for an assigned section.	Attendance is saved and reflected in the student's attendance history.
TC-06	Attendance	System computes attendance percentage after several recorded days.	Computed percentage matches manual calculation.
TC-07	Grades	Teacher encodes a quarterly grade for a subject.	Grade is saved and the quarterly average recalculates correctly.
TC-08	Inventory	Administrator reduces stock of an item below its reorder level.	System displays a low-stock notification.
TC-09	Messaging	Teacher sends a message regarding a specific student.	Message appears in the parent's message list linked to that student.
TC-10	Reports	Administrator exports an attendance summary report.	A correctly formatted CSV/Excel file is downloaded.
TC-11	Access Control	Parent account attempts to access the Administrator dashboard directly via URL.	Access is denied and the user is redirected to their assigned dashboard.
TC-12	Audit Log	Administrator deletes a user account.	Action is recorded in the audit log with actor, action, and timestamp.

Table L.1 — Representative Functional Test Cases

M

Sample Report Output Layouts

The following tables illustrate the intended layout of key exportable reports described in Section 3.9.1. Actual values shown are illustrative placeholders only.

Appendix

M.1 Student Masterlist Report (Layout)

Student ID	Name	Grade Level	Section	Status
0001	Sample, Student A.	Grade 4	Faith	Active
0002	Sample, Student B.	Grade 4	Faith	Active

M.2 Attendance Summary Report (Layout)

Student ID	Name	Days Present	Days Absent	Attendance %
0001	Sample, Student A.	85	5	94.4%
0002	Sample, Student B.	80	10	88.9%

M.3 Grade Report (Layout)

Student ID	Name	Subject	Quarter	Grade
0001	Sample, Student A.	Mathematics	Q1	88
0001	Sample, Student A.	English	Q1	90

M.4 Inventory Report (Layout)

Item	Quantity on Hand	Reorder Level	Status
Bond Paper (ream)	12	10	OK
Whiteboard Marker	3	5	Low Stock

N

Risk Management Plan

Identified Risk	Likelihood	Mitigation Strategy
Scope creep during development	Medium	Adhere to the requirements specification in Section 3.9; treat additional requests as recommendations for future work.
Delay in stakeholder availability for interviews/testing	Medium	Schedule data-gathering activities early, with buffer time built into the timeline in Appendix G.
Data loss during development or migration	Low	Use only anonymized or fictionalized sample data during development, with regular backups.
Unfamiliarity with a chosen tool or framework	Low	Allocate initial development weeks to a short technical ramp-up before core module development begins.
Resistance to adoption among end users	Medium	Apply the phased rollout and orientation approach discussed in Sections 1.7 and 3.11.2.

Appendix O

Maintenance and Sustainability Plan

Following deployment, routine maintenance of the system — including periodic backups, security updates to the underlying software stack, and minor bug fixes — is recommended to be assigned to a designated staff member or, during the academic year immediately following the capstone project, to the researchers under a transition agreement with the institution.

To ensure the long-term sustainability of the system beyond the researchers' direct involvement, the researchers recommend that the institution budget for continued web hosting on an annual basis, that at least one staff member receive basic administrator-level training beyond the orientation described in Appendix K, and that the source code and documentation — including the data dictionary in Appendix F — be formally turned over to the institution upon completion of Capstone 2, so that future student developers or hired contractors can maintain or extend the system without depending exclusively on the original researchers.

Recommended future enhancements, beyond the scope of the present study, include mobile application support, integration with the Department of Education's Learner Information System, and the introduction of predictive inventory analytics as the institution's scale grows, consistent with the possibilities noted in Sections 1.9 and 2.6.

P

Extended Discussion on Design Rationale and Alternatives Considered This appendix expands on the design decisions introduced in Chapter 3 by discussing, for each

Appendix

major design choice, the principal alternatives the researchers considered and the reasons those alternatives were set aside in favor of the approach ultimately adopted.

P.1 Attendance Identification Method

Three alternatives were considered for recording daily attendance: biometric fingerprint scanning, QR-code identification cards, and direct teacher encoding through a class list. Biometric scanning was set aside primarily on cost grounds, since dedicated scanner hardware for every classroom would exceed the institution's budget, and on practicality grounds, since very young learners in the lower grade levels are not always able to reliably operate a fingerprint scanner unassisted. QR-code identification was considered more affordable, but it introduces a dependency on students consistently carrying a physical card or a personal device capable of displaying a digital code, which is not guaranteed among all households served by the institution. Direct teacher encoding, in which the classroom teacher marks each student's status from a class list already open during the first period, was ultimately selected because it requires no additional hardware beyond the device the teacher already uses to access the system, and because it most closely mirrors the manual process already familiar to teachers, in keeping with the compatibility principle discussed under the Diffusion of Innovation Theory in Section 1.7.

P.2 Database Model

The researchers considered maintaining separate databases per module (a database for student records, a separate one for attendance, and so on) versus a single, unified relational

database in which all entities reference a common Students table. The separate-database approach was set aside because it would have recreated, in digital form, the same fragmentation problem the study seeks to solve — namely, that information about a single student is scattered across disconnected sources, as described in Section 1.1. The unified relational model, detailed in the data dictionary in Appendix F, was selected because it allows a single student record to be the anchor point from which attendance, grades, and messages are all derived, ensuring consistency and avoiding duplicate encoding.

P.3 User Interface Approach

A feature-rich, highly customizable dashboard was considered, in which users could rearrange widgets and choose from multiple report formats. This was set aside in favor of a simpler, fixed-layout dashboard per role, directly informed by the perceived-ease-of-use concern central to the Technology Acceptance Model discussed in Section 1.7. Because the intended end users include parents and staff with limited prior exposure to web-based systems, a smaller number of clearly labeled, non-configurable views was judged more likely to support sustained adoption than a more flexible but potentially confusing interface.

P.4 Hosting and Deployment Model

The researchers considered a self-hosted, on-premises server maintained by the institution versus a commodity, third-party web hosting arrangement. Self-hosting was set aside because it would require the institution to maintain dedicated hardware, ensure continuous power and connectivity, and assign staff to server administration — responsibilities disproportionate to the scale of a small private school. Commodity web hosting was selected because it shifts responsibility for hardware maintenance, power, and baseline security

Appendix

patching to the hosting provider, at a predictable and comparatively low recurring cost, consistent with the economic feasibility analysis in Section 3.11.3.

Q

Localized Context: ICT Readiness of Rural Philippine Private Learning Centers

This appendix situates the study within the broader context of information and communications technology (ICT) adoption among small, rural, private learning institutions in the Philippines, an environment noticeably different from the large, well-resourced urban universities more commonly represented in the school-management-system literature reviewed in Chapter 2.

Small private learning centers of the type represented by Tigbauan Fundamental Learning Center, Inc. typically operate on tuition revenue alone, without the subsidies or large endowments available to bigger institutions, which constrains the amount that can be allocated to information technology infrastructure in any given year. Reports on the broader Philippine basic education ecosystem note that school-level ICT readiness varies considerably even within the same municipality, depending heavily on the individual initiative of school leadership rather than on any uniform baseline of provided equipment (Cuyegkeng, 2026). This variability underscores why a general, one-size-fits-all commercial school management platform — often priced and designed for larger institutions — is frequently a poor fit for a school of this scale, and why a custom-built, appropriately scoped system such as the one proposed in this study can offer a more suitable alternative.

At the same time, rural connectivity in the Western Visayas region, where the institution is located, has improved substantially in recent years relative to previous decades, with mobile broadband coverage now reaching most municipalities, even if bandwidth and reliability still lag behind urban centers. This improving — if still imperfect — connectivity is what makes a

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web-based, rather than a purely offline, system a realistic proposition for the institution at this time, provided that the system is designed, as discussed in Table 3.1, to remain responsive under modest network conditions rather than assuming abundant bandwidth.

The researchers also note that digital literacy among staff and parents in this context tends to be uneven: while most adults in the community are familiar with basic smartphone use, particularly messaging and social media applications, fewer have prior experience with web-based administrative systems of the kind proposed here. This observation reinforces the emphasis, throughout Chapter 3, on interface simplicity and on a phased, well-supported rollout rather than an assumption of readily available technical sophistication among end users.

Taken together, these contextual factors — constrained budgets, uneven but improving connectivity, and variable digital literacy — do not argue against digitization; rather, they argue for a deliberately modest, carefully scoped system whose features are matched to what the institution can realistically sustain, which is precisely the design philosophy the researchers have sought to apply throughout this proposal.

R

Stakeholder Roles and Responsibilities During Development

Successful completion of the proposed system depends on the coordinated participation of several stakeholders beyond the researchers themselves. This appendix clarifies the expected role of each.

R.1 The Researchers (Proponents)

The four student-researchers are jointly responsible for all phases of the system's design, development, testing, and documentation described in Chapter 3, including the conduct of interviews and surveys described in Section 3.5, subject to the ethical safeguards discussed in Section 3.13.

R.2 The Capstone Adviser

The assigned capstone adviser is responsible for reviewing the researchers' progress at each major milestone identified in the timeline in Appendix G, providing guidance on the technical soundness of the system design, and endorsing the completed manuscript and prototype for panel evaluation.

R.3 The School Administrator and Staff of Tigbauan Fundamental Learning Center, Inc.

The institution's Administrator and teaching staff are expected to participate in the data-gathering activities described in Sections 3.5 and Appendix C and D, to provide access to anonymized or fictionalized sample records for development and testing purposes, and to participate in the user acceptance evaluation described in Section 3.7 once a working prototype is available.

R.4 The Capstone Panel

The evaluation panel is responsible for independently assessing the proposal and, at a later stage, the completed system against the institution's academic standards for capstone research, providing feedback that the researchers are expected to incorporate into subsequent revisions.

Appendix S

Data Migration Plan

Because the institution's existing records are entirely paper-based, as described in Section 1.1, a structured migration plan is necessary to transfer this information into the new system without introducing errors. The researchers propose the following approach, to be executed collaboratively with the Administrator during the initial deployment phase.

S.1 Batching

Records will be migrated in batches by grade level, beginning with the lowest enrollment grade level to allow the Administrator and researchers to refine the migration procedure on a smaller, more manageable set of records before proceeding to larger grade levels.

S.2 Data Entry and Verification

For each batch, student information will first be encoded into the system by designated staff, after which a second person — either another staff member or a researcher — will independently verify the encoded information against the original paper record, correcting any discrepancies before the batch is marked complete. This two-person verification step is intended to catch transcription errors before they propagate into attendance and grade records that depend on the underlying student record.

S.3 Historical Records

Given the time constraints of the capstone timeline, the researchers recommend that only the current school year's active student records be fully migrated during initial

deployment, with prior years' historical records digitized progressively thereafter as time permits, rather than attempting to migrate the institution's entire historical archive at once.

S.4 Parallel Run

During an initial transition period, the researchers recommend that the institution maintain its existing manual attendance and grade records alongside the new system, allowing staff to cross-check the system's output against familiar manual processes before fully retiring the paper-based method, consistent with the trialability principle discussed in Section 1.7.

Appendix T

Security and Access Control Policy

This appendix expands on the security-related non-functional requirements introduced in Table 3.1 by specifying the concrete policies the system is expected to enforce.

T.1 Password Policy

User passwords shall be required to meet a minimum length and complexity standard at the time of account creation, shall never be stored or transmitted in plain text, and shall be stored only as a one-way cryptographic hash, consistent with the security requirement in Table 3.1 and the data-protection obligations discussed in Section 2.5.

T.2 Session Management

User sessions shall automatically expire after a period of inactivity, requiring reauthentication before further access is granted, in order to reduce the risk of an unattended, logged-in device being used by an unauthorized person, particularly in shared administrative spaces.

T.3 Role-Based Access Control

Every function of the system shall be gated by the authenticated user's assigned role (Administrator, Teacher, or Parent), such that a Teacher account, for example, cannot access another teacher's section without explicit assignment, and a Parent account cannot view any student record other than that of their own linked child or children, as described in Section 3.9.1.

T.4 Audit Review

The Administrator shall be able to review the audit log described in Appendix F.9 on a periodic basis, allowing significant actions — such as the creation or deactivation of user accounts, or bulk edits to student records — to be traced to a specific user and timestamp, supporting accountability in the event of a data discrepancy.

Appendix U

Detailed Use Case Specifications

This appendix expands each core use case introduced in the use case diagram (Figure 3.3) into a full specification, describing the primary actor, preconditions, main flow of events, principal alternate flow, and postconditions. These specifications provided the detailed basis from which the functional requirements in Section 3.9.1 and the test cases in Appendix L were derived.

U.1 Log In

- **Primary Actor:** All registered users (Administrator, Teacher, Parent).
- **Preconditions:** The user has an active account created by the Administrator.

Main Flow:

1. The user navigates to the system login page.
2. The user enters their username and password.
3. The system verifies the submitted credentials against the stored password hash.
4. The system establishes an authenticated session and redirects the user to the dashboard corresponding to their role.

Alternate Flow:

- If the submitted credentials do not match a stored account, the system displays an invalid-credentials message and does not establish a session.
- **Postconditions:** The user is logged in and presented with the dashboard appropriate to their assigned role.

U.2 Manage Students

- **Primary Actor:** Administrator.

- **Preconditions:** The Administrator is logged in and has navigated to the Manage Students screen.

Main Flow:

- 1. The Administrator selects “Add New Student,” “Edit,” or “Deactivate” for an existing record.
- 2. For a new record, the Administrator completes the required fields (name, birthdate, grade level, section) and optionally uploads a photo.
- 3. The system validates that all required fields are completed and correctly formatted.
- 4. The system saves the record and displays a confirmation message.

Alternate Flow:

- If a required field is missing or invalid, the system displays a validation error and does not save the record until corrected.
- **Postconditions:** The student record is created, updated, or marked inactive in the Students table, and is immediately available to other modules (attendance, grades, messaging) that reference it.

U.3 Manage Users

- **Primary Actor:** Administrator.
- **Preconditions:** The Administrator is logged in and has navigated to the Manage Users screen.

Main Flow:

- 1. The Administrator selects “Add New User” and enters the user’s name and contact information.
- 2. The Administrator assigns the appropriate role (Teacher or Parent).
- 3. The system generates the account and an initial set of login credentials.

- 4. The Administrator communicates the initial credentials to the user through an appropriate channel.

Alternate Flow:

- If the Administrator instead selects an existing user, the system presents options to edit the user's details or deactivate the account.
- **Postconditions:** A new user account exists with the correct role assignment, or an existing account is updated or deactivated.

U.4 Link Parent to Student

- **Primary Actor:** Administrator.
- **Preconditions:** Both the parent account and the student record already exist in the system.

Main Flow:

- 1. The Administrator opens the relevant student's record.
- 2. The Administrator selects "Link Parent Account" and searches for the parent's account by name.
- 3. The Administrator selects the correct account and confirms the relationship type.
- 4. The system creates the link and grants the parent account view access to that student's attendance and grade records.

Alternate Flow:

- If more than one parent account is associated with the same household, the Administrator repeats the process for each account.
- **Postconditions:** The parent account is linked to the correct student record and the corresponding entry exists in the Parent-Student Link table (Appendix F.2).

- **Preconditions:** The

U.5 Take Attendance

- **Primary Actor:** Teacher.

Teacher is logged in and assigned to at least one section.

Main Flow:

- 1. The Teacher selects “Attendance,” then the relevant section and date.
- 2. The system displays the class list for that section.
- 3. The Teacher marks each student’s status (Present, Absent, Late, or Excused).
- 4. The Teacher clicks “Save Attendance.”
- 5. The system stores the attendance record and updates the student’s computed attendance percentage.

Alternate Flow:

- If attendance for that section and date was already recorded, the system loads the existing entries for editing rather than creating duplicate records.
- **Postconditions:** Attendance for the selected section and date is recorded and reflected in each student’s attendance history and computed percentage.

U.6 Encode Grades

- **Primary Actor:** Teacher.
- **Preconditions:** The Teacher is logged in and assigned to teach the relevant subject and section.

Main Flow:

- 1. The Teacher selects “Grades,” then the relevant subject, section, and quarter.
- 2. The system displays the list of enrolled students for that subject and section.
- 3. The Teacher enters each student’s grade for the quarter.

-
- 4. The Teacher clicks “Save Grades.”
- 5. The system stores the grade records and recalculates the affected students’ quarterly averages.

Preconditions: The

Alternate Flow:

- If a previously encoded grade is being corrected, the system overwrites the prior value and logs the change in the audit log.
- **Postconditions:** Grades for the selected subject, section, and quarter are recorded, and quarterly averages are updated accordingly.

U.7 Manage Inventory

- **Primary Actor:** Administrator.
- **Preconditions:** The Administrator is logged in and has navigated to the Manage Inventory screen.

Main Flow:

- 1. The Administrator selects “Add Item” to register a new item, or selects an existing item to record a stock movement.
- 2. The Administrator enters the quantity added or removed and, where applicable, a reason.
- 3. The system updates the recorded stock level and appends a new entry to the inventory transaction log.
- 4. If the resulting stock level falls below the configured reorder level, the system displays a low-stock notification.

Alternate Flow:

- If the Administrator attempts to remove more stock than is currently on hand, the system displays a validation warning before allowing the transaction to proceed.
- **Postconditions:** The inventory item’s stock level and transaction history are updated to reflect the recorded movement.

U.8 Send and Receive Messages

- **Primary Actor:** Teacher, Parent, or Administrator.
- sender and intended recipient both have active accounts, and, where applicable, are linked to the relevant student.

Main Flow:

- 1. The sender opens the relevant student's profile or the messaging tab.
- 2. The sender selects the intended recipient and composes the message.
- 3. The sender clicks "Send."
- 4. The system stores the message and makes it visible in the recipient's message list.

Alternate Flow:

- If the sender is a Parent, the list of possible recipients is limited to the teacher(s) and Administrator associated with their linked student.
- **Postconditions:** The message is recorded and associated with the relevant student, sender, and recipient.

U.9 View Attendance and Grades (Parent)

- **Primary Actor:** Parent.
- **Preconditions:** The Parent is logged in and linked to at least one student record.

Main Flow:

- 1. The Parent selects the relevant child, if more than one is linked to the account.
- 2. The Parent selects "Attendance" or "Grades."
- 3. The system retrieves and displays the corresponding records for the selected child only.

Alternate Flow:

Preconditions: The

- If the Parent account is linked to more than one child, the system presents a selector before displaying records.
- **Postconditions:** The Parent views accurate, up-to-date attendance or grade information limited strictly to their own linked child or children.

U.10 Generate Reports

- **Primary Actor:** Administrator.
- **Preconditions:** The Administrator is logged in and relevant data (student, attendance, grade, or inventory records) exists in the system.

Main Flow:

- 1. The Administrator selects “Generate Reports” and chooses the desired report type.
- 2. The Administrator selects the relevant date range or grading period.
- 3. The system compiles the requested data and generates a downloadable file.
- 4. The Administrator clicks “Export” to download the file in CSV or Excel format.

Alternate Flow:

- If no records match the selected criteria, the system displays a message indicating that no data is available for export.
- **Postconditions:** A correctly formatted report file is generated and downloaded, suitable for institutional record-keeping or external submission.

Appendix V

Extended Literature: School Digitization in Resource-Constrained Settings

To further contextualize the present study, the researchers reviewed additional literature on the digitization of school administrative processes in resource-constrained settings beyond the Philippines, which offers useful points of comparison for a small private institution such as Tigbauan Fundamental Learning Center, Inc.

Studies of school-management-system adoption in other Southeast Asian and South Asian contexts consistently report that the single greatest barrier to sustained use is not the initial development or purchase of the system but the availability of ongoing technical support after the original developers or vendors move on, a challenge that closely parallels the sustainability concerns addressed in Appendix O of this proposal. Institutions that paired a new system with a clear, written maintenance and handover plan — rather than relying on the informal availability of the original developers — were more likely to report continued use of the system three or more years after initial deployment.

A recurring theme in this broader literature is the importance of designing for intermittent connectivity rather than assuming constant, high-speed internet access. Systems that degrade gracefully under poor connectivity — for instance, by allowing a teacher to complete attendance encoding even if a save operation must briefly retry due to a weak signal — were found to sustain higher user satisfaction than systems that simply failed or timed out under the same conditions. This consideration is reflected in the performance requirement specified in Table 3.1, which sets a modest, achievable page-load expectation rather than assuming ideal network conditions.

Comparative studies further note that institutions transitioning from fully manual processes benefit from an incremental rollout strategy — introducing one module at a time, beginning with the function causing the greatest administrative burden — rather than attempting a simultaneous, institution-wide switch to a fully digital process. This finding is consistent with the phased approach recommended throughout Chapter 3 and Appendix S of this proposal, in which student records and attendance are prioritized ahead of inventory and messaging.

Finally, the reviewed literature emphasizes that perceived data security is often as important to user trust as actual technical security measures: parents and staff are considerably more willing to adopt a digital system when the institution is able to clearly explain, in plain language, how student data is protected and who can access it. This observation reinforces the value of the security and access control policy detailed in Appendix T, and suggests that this policy should be communicated to parents in an accessible form — rather than only documented internally — as part of the system’s eventual rollout.

Appendix W

Comparison of Alternative System Development Methodologies

Section 3.2 identifies the Agile Development Model as the methodology adopted for this study. This appendix documents the comparison the researchers undertook among the methodologies considered before arriving at that decision.

W.1 Waterfall Model

The Waterfall Model organizes development into strictly sequential phases — requirements, design, implementation, testing, and maintenance — with each phase expected

to be completed before the next begins. Its principal advantage for a capstone project is the clarity of its milestones, which map neatly onto a fixed academic timeline. However, the researchers judged this rigidity to be a significant drawback given that the requirements for a first-time client institution such as Tigbauan Fundamental Learning Center, Inc. were expected to be refined through direct observation of early prototypes rather than fully specified up front. Committing to a complete, unchangeable requirements document before any development began would have risked producing a system that technically satisfied its initial specification but failed to reflect adjustments that only became apparent once the Administrator and teachers could interact with a working prototype.

W.2 Scrum

Scrum, a specific Agile framework built around fixed-length sprints, dedicated ceremonies (sprint planning, daily stand-ups, sprint review, and retrospective), and defined roles such as Scrum Master and Product Owner, was also considered. While Scrum shares the iterative philosophy the researchers ultimately adopted, its full ceremonial structure was judged to be heavier than necessary for a four-person student team operating without a dedicated product owner role distinct from the researchers themselves. The researchers instead adopted the general iterative and incremental spirit of Agile, described in Section 3.2, without imposing the complete formal Scrum ceremony set, allowing the team to adapt sprint length and review frequency to the academic calendar.

W.3 Rapid Application Development (RAD)

Rapid Application Development emphasizes fast, prototype-driven development with heavy end-user involvement and a de-emphasis on rigid upfront planning. RAD's emphasis on early, tangible prototypes appealed to the researchers, particularly given the trialability

principle discussed in Section 1.7. However, RAD's traditional reliance on highly available end users for continuous feedback throughout development was judged difficult to guarantee in this context, since the Administrator and teachers at Tigbauan Fundamental Learning Center, Inc. have their own demanding daily responsibilities and cannot be expected to participate in development-cycle-length feedback sessions as frequently as classic RAD assumes. The researchers instead incorporated RAD's emphasis on early prototyping within the broader Agile structure, scheduling feedback checkpoints at the end of each development phase rather than continuously throughout.

W.4 Rationale for the Chosen Approach

The Agile Development Model, as applied in Section 3.2, was ultimately selected because it preserves the ability to adjust requirements and interface design based on stakeholder feedback gathered at defined checkpoints, without imposing either the rigidity of Waterfall or the continuous end-user availability assumed by classic RAD. This makes it, in the researchers' judgment, the methodology best matched to the realistic constraints of both the development team and the client institution.

Appendix X

Detailed Description of Dashboard Screens

This appendix describes, in narrative form, the intended layout and content of each rolebased dashboard referenced throughout Chapter 3 and Appendix K, to guide the interface development undertaken in Capstone 2.

X.1 Administrator Dashboard

Upon logging in, the Administrator is presented with a summary view showing the total number of active students, a count of any pending low-stock inventory alerts, and a short list of recent audit log entries. A persistent side navigation menu provides access to Manage Students, Manage Users, Manage Inventory, Generate Reports, and View Audit Log, corresponding to the use cases detailed in Appendix U. The summary view is intended to give the Administrator, at a glance, a sense of whether anything requires immediate attention — such as a low-stock item — without requiring navigation into a deeper screen.

X.2 Teacher Dashboard

The Teacher dashboard opens to a view of the teacher's assigned sections and subjects, with quick links to Take Attendance and Encode Grades for the current date or active grading period. A separate Messages tab lists recent messages exchanged with parents, organized by student, so that a teacher can quickly locate the conversation history relevant to a particular learner. The dashboard deliberately omits any function outside the Teacher's assigned classes, consistent with the role-based access control policy detailed in Appendix T.3.

X.3 Parent Dashboard

The Parent dashboard opens to a simple selector for the linked child or children, followed by three clearly labeled views: Attendance, Grades, and Messages. Each view presents only the information relevant to the selected child, formatted in plain language consistent with the usability requirement in Table 3.1, and avoiding administrative terminology that may be unfamiliar to a parent audience. The Messages view allows the parent to initiate or continue a conversation with the relevant teacher or the Administrator, consistent with use case U.8 in Appendix U.

Appendix Y

Glossary of Technical Terms and Acronyms

The following glossary supplements the Definition of Terms in Section 1.6 with technical terms and acronyms used throughout the systems-analysis discussion in Chapter 3 and its appendices.

- **LRN:** Learner Reference Number, a unique identifier issued by the Department of Education to a learner within the Philippine basic education system.
- **RBAC:** Role-Based Access Control, a security approach that restricts system access according to a user's assigned role rather than granting access on an individual basis.
- **CRUD:** Create, Read, Update, Delete — the four basic operations supported by most data-management interfaces, including the Manage Students and Manage Users screens described in Section 3.9.
- **DFD:** Data Flow Diagram, a graphical representation of how data moves through a system, illustrated for this study in Figure 3.1.
- **ERD:** Entity Relationship Diagram, a graphical representation of a system's data entities and their relationships, illustrated for this study in Figure 3.2 and detailed in the data dictionary in Appendix F.
- **SRS:** Software Requirements Specification, the consolidated statement of functional and non-functional requirements presented in Section 3.9.
- **UI/UX:** User Interface / User Experience, referring respectively to the visual design of a system's screens and to the overall quality of a user's interaction with the system.
- **API:** Application Programming Interface, a defined mechanism through which separate software components or systems communicate with one another.
- **CSV:** Comma-Separated Values, a plain-text file format commonly used for exporting tabular data such as the reports described in Section 3.9.1.

- **UAT:** User Acceptance Testing, the evaluation stage, described in Section 3.7, in which representative end users assess whether the completed system meets their practical needs.

Appendix Z

Detailed Discussion of Evaluation Criteria and Instrument

Section 3.7 identifies the general evaluation criteria used to assess the completed prototype. This appendix expands that discussion by aligning the study's evaluation approach with the internationally recognized ISO/IEC 25010 software product quality model, and by presenting the evaluation instrument to be administered to representative end users during the user acceptance evaluation.

Z.1 Functional Suitability

Functional suitability refers to the degree to which the system provides functions that meet stated and implied needs when used under specified conditions. In this study, functional suitability is assessed against the functional requirements enumerated in Section 3.9.1 and the use case specifications in Appendix U, through the functional test cases documented in Appendix L. A system is considered to satisfy this criterion when each core function — managing student records, recording attendance, encoding grades, tracking inventory, and exchanging messages — produces the expected result described in its corresponding test case.

Z.2 Performance Efficiency

Performance efficiency concerns the amount of resources used and the response time exhibited by the system under stated conditions. Consistent with the performance requirement in Table 3.1, this study evaluates performance efficiency primarily through observed page-load times for common operations — opening the dashboard, submitting attendance for a section, and generating a report — under the ordinary network conditions available at the institution, rather than under laboratory conditions that may not reflect realworld use.

Z.3 Compatibility

Compatibility refers to the degree to which the system can exchange information with other systems and perform its required functions while sharing a common environment with other software. For this study, compatibility is assessed primarily in terms of browser and device compatibility — whether the system functions correctly across the common desktop and mobile browsers available to the institution’s staff and parents — rather than integration with external third-party systems, which is treated as a future enhancement in Appendix O.

Z.4 Usability

Usability, closely related to the Technology Acceptance Model discussed in Section 1.7, refers to the degree to which the system can be used by specified users to achieve specified goals with effectiveness, efficiency, and satisfaction. This study evaluates usability through direct observation of representative users completing core tasks (for example, a teacher taking attendance for the first time without extensive guidance) and through the survey instrument described below, which asks users to rate their experience using the system.

Z.5 Reliability

Reliability refers to the degree to which the system performs specified functions under specified conditions for a specified period of time without failure. In this study, reliability is assessed through repeated execution of the same operation (for example, encoding the same set of grades multiple times under simulated conditions) to confirm consistent, correct results, and through observation of the system’s behavior when a save operation is interrupted, to confirm that partial or corrupted data is not written to the database.

Z.6 Security

Security, detailed further in Appendix T, refers to the degree to which the system protects information and data so that unauthorized persons or systems cannot access, modify, or destroy it. This study evaluates security through the access-control test cases in Appendix L (for example, TC-11), confirming that a user cannot access functions or data outside those permitted by their assigned role.

Z.7 Maintainability

Maintainability refers to the degree of effectiveness and efficiency with which the system can be modified by its intended maintainers. This study evaluates maintainability qualitatively, through review of the code organization described in Table 3.1 and the completeness of the documentation — including the data dictionary in Appendix F and the user manual in Appendix K — that would be available to a future maintainer who was not part of the original development team.

Z.8 Portability

Portability refers to the degree of effectiveness and efficiency with which the system can be transferred from one hardware, software, or operational environment to another.

Because the system is web-based and hosted on commodity infrastructure, as discussed in Appendix P.4, portability is assessed primarily in terms of the ease with which the system's database and application code could be migrated to an alternative hosting provider, should the institution need to change providers in the future.

Z.9 Evaluation Instrument

Table Z.1 presents the evaluation instrument to be administered to the Administrator, participating teachers, and participating parents during the user acceptance evaluation described in Section 3.7, rated on the same four-point scale used in Appendix C.

Quality Characteristic	Evaluation Statement
Functional Suitability	The system performs the functions I need without errors.
Performance Efficiency	The system responds quickly enough for my daily tasks.
Compatibility	The system works properly on the device and browser I normally use.
Usability	The system is easy to learn and use without extensive assistance.
Reliability	The system consistently produces correct, consistent results.
Security	I am confident that my access is appropriately limited to what I need.
Maintainability	(Rated by Administrator/technical evaluators only) The system and its documentation appear well organized.
Portability	(Rated by technical evaluators only) The system does not appear tied to a single, irreplaceable hosting setup.

Table Z.1 — User Acceptance Evaluation Instrument (ISO/IEC 25010-Aligned)

Results gathered through this instrument, together with the functional, performance, and security test results documented in Appendix L, will together form the basis of the system evaluation report to be completed in Capstone 2.

Appendix

AA

Institutional Profile and Alignment with Institutional Goals

This appendix provides additional background on Tigbauan Fundamental Learning Center, Inc. and explains how the proposed system aligns with the institution's broader operational goals, complementing the background discussion in Section 1.1.

As a small, private basic-education institution serving the municipality of Tigbauan and its surrounding communities, the school's core mission centers on providing accessible, quality basic education to learners in the local community, supported by a lean administrative staff that, by necessity, handles multiple responsibilities beyond direct instruction. This operational reality — a small staff performing enrollment, records-keeping, attendance monitoring, grading, inventory oversight, and parent communication with limited dedicated administrative personnel — is precisely the condition that motivates the integrated, singleplatform approach adopted in this study rather than a collection of separate specialized tools, each of which would impose its own separate learning curve on an already stretched staff.

The proposed system is intended to support, rather than replace, the institution's existing pedagogical priorities. Time currently spent by teachers on manual grade computation and attendance tallying, and by the Administrator on manually cross-referencing paper records, represents time not spent on direct instructional support, mentoring, or parent engagement. By automating the more mechanical aspects of these tasks — as detailed in the functional requirements in Section 3.9.1 — the system is designed to return time to staff for the activities more directly aligned with the institution's educational mission.

The researchers also note that the institution's continued growth — whether through increasing enrollment or the eventual addition of new grade levels or programs — is likely to make the existing manual processes progressively less sustainable, since the administrative burden described in Section 1.1 tends to scale with enrollment while the size of the administrative staff does not necessarily scale at the same rate. The scalability requirement specified in Table 3.1, which calls for a database design that accommodates growth in enrollment and historical records without requiring a schema redesign, is intended specifically to ensure that the system remains a viable long-term investment for the institution rather than a short-term convenience that must be replaced as the school grows.

Finally, the researchers observe that the institution's standing within its community depends in part on the confidence parents place in its ability to responsibly safeguard their children's records and to communicate clearly and promptly regarding their children's attendance and academic performance. The security and access control measures detailed in Appendix T, and the parent-facing communication features detailed in Section 3.9.1 and Appendix X.3, are accordingly presented not merely as technical features but as contributions to the institution's broader relationship of trust with the families it serves.

In this sense, the proposed system is framed throughout this proposal not as a standalone technical exercise but as a tool in service of the institution's continued ability to fulfill its educational mission efficiently, securely, and in a manner that supports rather than burdens its teaching staff.

Appendix BB

Summary of Recommendations for Capstone 2

This final appendix consolidates the recommendations distributed throughout the preceding sections and appendices into a single reference list to guide the researchers' work during Capstone 2.

- 1. Finalize and pilot-test the survey and interview instruments in Appendices C and D with a small sample before full administration.
- 2. Complete the detailed, week-by-week Gantt chart in Appendix G with specific task owners once the four researchers finalize their individual role assignments.
- 3. Develop the student records, attendance, and grades modules first, consistent with the phased rollout strategy discussed in Appendix S, before proceeding to inventory and messaging.
- 4. Execute the full set of functional, performance, and security test cases outlined in Appendix L, expanding the representative subset shown into a complete test suite.
- 5. Capture actual screenshots of the finalized interface to replace the narrative descriptions in Appendices K and X.
- 6. Administer the ISO/IEC 25010-aligned evaluation instrument in Appendix Z to the Administrator, participating teachers, and participating parents once a stable prototype is available.
- 7. Finalize the cost figures in Table 3.3 based on the actual hosting arrangement selected, replacing the indicative estimates provided.
- 8. Formalize the data migration procedure in Appendix S with the Administrator, including a specific schedule for migrating each grade level.

- 9. Prepare the final turnover documentation — including source code, the data dictionary in Appendix F, and the user manual in Appendix K — for handover to the institution as discussed in Appendix O.
- 10. Revisit the risk register in Appendix N at each major milestone to confirm that identified risks remain adequately mitigated or to add newly identified risks.

Taken together, these recommendations are intended to carry the work documented in this proposal through to a fully developed, tested, and evaluated system by the conclusion of Capstone 2, consistent with the objectives set out in Section 1.3.

Appendix CC

Closing Notes

The researchers wish to note that the content of this proposal, including the appendices, represents a good-faith projection of the design, methodology, and supporting documentation expected to be developed and validated in full during Capstone 2. Instruments described as templates (Appendices B, C, D, and H), layouts described as illustrative (Appendices E and M), and estimates described as indicative (Table 3.3 and the Gantt chart in Appendix G) are explicitly identified as such throughout this document and are intended to be finalized, executed, and, where applicable, replaced with actual gathered data as the project progresses.

The researchers further acknowledge the guidance of their capstone adviser and the cooperation of the administration of Tigbauan Fundamental Learning Center, Inc. in the preparation of this proposal, and look forward to their continued partnership through the completion of the proposed system.

This concludes the proposal document.

Appendix DD

Additional Glossary Terms

The following terms supplement the glossary in Appendix Y.

- **Agile:** An iterative approach to software development that delivers work in small, incremental cycles rather than a single, large release, as discussed in Section 3.2 and Appendix W.
- **Prototype:** A working, but not necessarily complete, early version of a system used to gather feedback before full development is finished, referenced throughout Appendix P and W.
- **Stakeholder:** Any individual or group with an interest in the outcome of the project, including the Administrator, teachers, parents, and the capstone panel, as enumerated in Appendix R.
- **Reorder Level:** The minimum stock quantity at which the system triggers a lowstock notification, as defined in the Inventory table in Appendix F.6.
- **Session:** A period of authenticated interaction between a user and the system, beginning at login and ending at logout or timeout, as discussed in Appendix T.2.
- **Hash:** A one-way transformation applied to a password so that the original password cannot be recovered from the stored value, as discussed in Appendix T.1.
- **Masterlist:** A consolidated list of all currently enrolled students, exportable as a report per Section 3.9.1 and illustrated in Appendix M.1.

Appendix

- **Turnover:** The formal handover of the completed system, its source code, and its documentation to the client institution, as discussed in Appendix O.

EE

Note on Document Scope

This proposal document, comprising Chapters 1 through 3 and the accompanying appendices, represents the complete Capstone 1 deliverable. Chapters 4 and 5 — covering presentation of development results, discussion of findings, and the study’s final summary, conclusions, and recommendations — fall outside the scope of Capstone 1 and will be prepared during Capstone 2, once the system described herein has been fully developed, tested, and evaluated in accordance with the plan set out in Chapter 3 and its appendices.

The researchers have organized the appendices to mirror, as closely as practicable, the eventual structure of the full capstone manuscript, so that material such as the data dictionary (Appendix F), the user manual (Appendix K), and the evaluation instrument (Appendix Z) can be carried forward with minimal restructuring once actual development results are available to populate them.

FF

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